

# Statistical Inference for Linear Stochastic Approximation with Richardson-Romberg Extrapolation

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## Abstract

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## 1. Introduction

Stochastic approximation (SA) algorithms are a cornerstone of modern computational statistics, optimization, and reinforcement learning. Introduced by Robbins and Monro (1951), these iterative procedures provide a principled way to find roots of equations or optimize objectives when only noisy observations are available. A particularly important subclass is the *linear stochastic approximation* (LSA) algorithm, which arises naturally in temporal-difference (TD) learning, policy evaluation, and stochastic gradient descent for linear models.

In this work, we study the LSA recursion with a *constant step size*  $\alpha > 0$ :

$$\theta_k^{(\alpha)} = \theta_{k-1}^{(\alpha)} - \alpha \{A(Z_k)\theta_{k-1}^{(\alpha)} - b(Z_k)\}, \quad k \geq 1,$$

where  $\{Z_k\}_{k \in \mathbb{N}}$  is a time-homogeneous Markov chain on a measurable space  $(Z, \mathcal{Z})$  with transition kernel  $Q$  and unique invariant distribution  $\pi$ . The mappings  $A : Z \rightarrow \mathbb{R}^{d \times d}$  and  $b : Z \rightarrow \mathbb{R}^d$  are measurable functions satisfying  $\bar{A} := \int_Z A(z) d\pi(z)$  and  $\bar{b} := \int_Z b(z) d\pi(z)$ . Assuming that  $-\bar{A}$  is a *Hurwitz matrix* (all eigenvalues have strictly negative real parts), the target parameter  $\theta^* = \bar{A}^{-1}\bar{b}$  is uniquely defined.

The *Polyak–Ruppert averaging* procedure (Polyak, 1990; Ruppert, 1988) provides an effective variance reduction technique. Given a burn-in period  $n_0 \geq 0$ , the averaged iterate is defined as

$$\bar{\theta}_n^{(\alpha)} = \frac{1}{n - n_0} \sum_{k=n_0}^{n-1} \theta_k^{(\alpha)}.$$

### 1.1. Bias of constant step-size iterates

The use of a constant step size  $\alpha > 0$  offers several practical advantages: it enables geometrically fast forgetting of the initial condition (Dieuleveut, Durmus, and Bach, 2020) and simplifies hyperparameter tuning compared to diminishing step-size schedules. However, unlike the classical regime  $\alpha_k \rightarrow 0$  with  $\sum \alpha_k = \infty$  and  $\sum \alpha_k^2 < \infty$ , a constant step size produces iterates that converge only *in distribution* to a stationary measure  $\Pi_\alpha$ , rather than almost surely to  $\theta^*$ . The stationary expectation  $\mathbb{E}[\theta_\infty^{(\alpha)}]$  is generally *biased* with respect to  $\theta^*$ , and this bias cannot be eliminated by Polyak–Ruppert averaging alone.

As shown in Levin, Naumov, and Samsonov (2025) and Huo, Chen, and Xie (2023), the asymptotic bias admits a power-series expansion in  $\alpha$ :

$$\lim_{n \rightarrow \infty} \mathbb{E}[\theta_n^{(\alpha)}] = \theta^* + \alpha \Delta + O(\alpha^2),$$

where  $\Delta = \bar{A}^{-1} \sum_{k=1}^{\infty} \mathbb{E}[\{Q^k \tilde{A}(Z_{\infty})\} \epsilon(Z_{\infty})]$  depends on the correlation structure of the Markov chain, and  $\tilde{A}(z) = A(z) - \bar{A}$  is the centered matrix-valued function.

## 1.2. Richardson–Romberg extrapolation

To eliminate the leading  $O(\alpha)$  bias term, we employ the *Richardson–Romberg* (RR) *extrapolation* procedure. Two LSA sequences are run *on the same Markov chain trajectory*  $\{Z_k\}$  with step sizes  $\alpha$  and  $2\alpha$ , and the RR iterate is formed as

$$\bar{\theta}_n^{(\alpha, \text{RR})} = 2\bar{\theta}_n^{(\alpha)} - \bar{\theta}_n^{(2\alpha)}.$$

Since both sequences share the same noise realization, the leading bias term  $\alpha\Delta$  cancels, leaving a residual bias of order  $O(\alpha^{3/2})$  or higher (Levin et al., 2025).

More generally, one can consider the multi-level extrapolation with  $M$  step sizes  $\mathcal{A} = \{\alpha_1, \dots, \alpha_M\}$  and coefficients  $\{h_m\}$  determined by the Vandermonde system (Huo et al., 2023):

$$\sum_{m=1}^M h_m = 1, \quad \sum_{m=1}^M h_m \alpha_m^l = 0, \quad l = 1, \dots, M-1,$$

which reduces the bias from  $O(\alpha)$  to  $O(\alpha^M)$ .

## 1.3. Problem statement and goals

The high-order moment bounds for the RR iterate  $\bar{\theta}_n^{(\alpha, \text{RR})}$  established in Levin et al. (2025) show that the leading error term scales as  $\sqrt{\text{Tr } \Sigma_{\epsilon}^{(M)}} \cdot n^{-1/2}$ , where  $\Sigma_{\epsilon}^{(M)}$  is the Markovian noise covariance defined in Section 1.4 below, matching the minimax optimal rate. Berry–Esseen type bounds and bootstrap inference procedures for the *standard* Polyak–Ruppert average  $\bar{\theta}_n$  (without extrapolation) under Markovian noise have been obtained in Samsonov, Sheshukova, Moulines, and Naumov (2025).

However, the *distributional approximation* — specifically, the central limit theorem and Berry–Esseen bounds — for the *averaged Richardson–Romberg iterates*  $\bar{\theta}_n^{(\alpha, \text{RR})}$  remains an open problem. The main goal of this work is to fill this gap by establishing:

1. A *central limit theorem* for  $\sqrt{n}(\bar{\theta}_n^{(\alpha, \text{RR})} - \theta^*)$ , identifying the limiting covariance matrix.
2. *Non-asymptotic Berry–Esseen bounds* for the rate of convergence to the Gaussian limit.
3. A practical *inference procedure* (confidence intervals) based on the RR-extrapolated iterates.

## 1.4. Setting and assumptions

We now formalize the setting and state the assumptions that will be used throughout this work.

Let  $\{Z_k\}_{k \in \mathbb{N}}$  be a Markov chain on a complete separable metric space  $(Z, \mathcal{Z})$  with transition kernel  $Q$ .

**Assumption 1 (Uniform geometric ergodicity).** The kernel  $Q$  admits a unique invariant distribution  $\pi$  and is *uniformly geometrically ergodic*: there exists  $t_{\text{mix}} \in \mathbb{N}^*$  such that for all  $k \in \mathbb{N}^*$ ,

$$\Delta(Q^k) := \sup_{z, z' \in Z} \frac{1}{2} \|Q^k(z, \cdot) - Q^k(z', \cdot)\|_{\text{TV}} \leq (1/4)^{\lfloor k/t_{\text{mix}} \rfloor}.$$

Equivalently, there exist constants  $\zeta > 0$  and  $\rho \in (0, 1)$  such that  $\sup_z \|Q^k(z, \cdot) - \pi\|_{\text{TV}} \leq \zeta \rho^k$  for all  $k \geq 1$ .

**Assumption 2 (Hurwitz condition and boundedness).** The matrix  $-\bar{A}$  is Hurwitz, i.e., all eigenvalues of  $\bar{A}$  have strictly positive real parts. Moreover,

$$C_A := \max \left( \sup_{z \in Z} \|A(z)\|, \sup_{z \in Z} \|\tilde{A}(z)\| \right) < \infty,$$

where  $\tilde{A}(z) := A(z) - \bar{A}$ .

**Assumption 3 (Noise regularity).** The noise function  $\epsilon(z) = \tilde{A}(z)\theta^* - \tilde{b}(z)$ , where  $\tilde{b}(z) = b(z) - \bar{b}$ , satisfies

$$\|\epsilon\|_\infty := \sup_{z \in Z} \|\epsilon(z)\| < +\infty.$$

Under Assumptions 1–3, the error  $\theta_k^{(\alpha)} - \theta^*$  satisfies the recursion

$$\theta_k^{(\alpha)} - \theta^* = (I - \alpha A(Z_k))(\theta_{k-1}^{(\alpha)} - \theta^*) - \alpha \epsilon(Z_k).$$

## 1.5. Key quantities

The *Markovian noise covariance matrix* captures both the marginal variance and the temporal correlations of the noise:

$$\Sigma_\epsilon^{(M)} = \mathbb{E}_\pi [\epsilon(Z_0)\epsilon(Z_0)^\top] + 2 \sum_{\ell=1}^{\infty} \mathbb{E}_\pi [\epsilon(Z_0)\epsilon(Z_\ell)^\top].$$

This matrix is the limiting covariance in the Markov chain CLT for the partial sums  $n^{-1/2} \sum_{t=0}^{n-1} \epsilon(Z_t)$  (cf. Douc et al., 2018, Theorem 21.2.10).

The *asymptotically optimal covariance matrix* is given by

$$\Sigma_\infty = \bar{A}^{-1} \Sigma_\epsilon^{(M)} (\bar{A}^{-1})^\top.$$

This covariance is optimal in the Hájek–Le Cam sense and represents the best achievable asymptotic variance for estimating  $\theta^*$  via averaged LSA.

The *Lyapunov equation* plays a central role in the contraction analysis. For any  $P = P^\top \succ 0$ , there exists a unique  $Q = Q^\top \succ 0$  satisfying  $\bar{A}^\top Q + Q \bar{A} = P$ . Defining  $a = \lambda_{\min}(P)/(2 \|Q\|)$  and  $\kappa_Q = \lambda_{\max}(Q)/\lambda_{\min}(Q)$ , the key contraction property holds: for all  $\alpha \in [0, \alpha_\infty]$ ,

$$\|I - \alpha \bar{A}\|_Q^2 \leq 1 - \alpha a.$$

Since the iterates  $\theta_k^{(\alpha)}$  alone are generally not Markovian (due to the Markovian noise), we consider the *joint process*  $\left(\theta_k^{(\alpha)}, Z_{k+1}\right)$  with kernel

$$\bar{P}_\alpha f(\theta, z) = \int_Z Q(z, dz') f(F_{z'}(\theta), z'),$$

where  $F_z(\theta) = (I - \alpha A(z))\theta + \alpha b(z)$ . Under Assumptions 1–3, this joint chain admits a unique invariant distribution  $\Pi_\alpha$  for sufficiently small  $\alpha > 0$  (Levin et al., 2025).

## 2. Zeroth-Order Richardson–Romberg Difference

### 2.1. LSA Error Decomposition

We consider the recursion

$$\theta_k = \theta_{k-1} - \alpha_k (A(Z_k)\theta_{k-1} - b(Z_k)), \quad \alpha_k = \alpha = \text{const.}$$

Define the transition products

$$\Gamma_{m:k} = \prod_{l=m}^k (I - \alpha A(Z_l)).$$

We also introduce

$$J_k^{(0)} = - \sum_{l=1}^k \alpha \Gamma_{l+1:k} \epsilon(Z_l), \quad H_k^{(0)} = - \sum_{l=1}^k \alpha \Gamma_{l+1:k} (A(Z_l) - \bar{A}) J_{l-1}^{(0)},$$

with initial conditions  $J_0^{(0)} = H_0^{(0)} = 0$ .

A standard PR-averaged decomposition (cf. Section 4.1 below for the full derivation) yields

$$\sqrt{n}(\bar{\theta}_n^{(\alpha)} - \theta^*) = W + D_1,$$

where the leading martingale-like term is

$$W = - \frac{1}{\sqrt{n}} \sum_{l=1}^{n-1} Q_l \epsilon(Z_l), \quad Q_l = \alpha \sum_{k=l}^{n-1} (I - \alpha \bar{A})^{n-k-1},$$

and the residual term is

$$D_1 = \frac{1}{\sqrt{n}} \sum_{k=0}^{n-1} \Gamma_{1:k} (\theta_0 - \theta^*) + \frac{1}{\sqrt{n}} \sum_{k=1}^{n-1} J_k^{(0)}.$$

### 2.2. RR Combination and the $\tilde{J}_n^{(0,\alpha)}$ Term

Applying the decomposition of the previous subsection separately to step sizes  $\alpha$  and  $2\alpha$  and forming the Richardson–Romberg combination, we obtain

$$\begin{aligned} \theta_n^{(\text{RR},\alpha)} - \theta^* &= \left[ 2\Gamma_{1:n}^{(\alpha)} - \Gamma_{1:n}^{(2\alpha)} \right] (\theta_0 - \theta^*) \\ &\quad + \left[ 2J_n^{(0,\alpha)} - J_n^{(0,2\alpha)} \right] + \left[ 2J_n^{(1,\alpha)} - J_n^{(1,2\alpha)} \right] + \left[ 2H_n^{(1,\alpha)} - H_n^{(1,2\alpha)} \right]. \end{aligned}$$

The first bracket is the deterministic transient and is exponentially small. The remaining three brackets are stochastic and need separate analysis. In this subsection we focus on the zeroth-order RR difference, which we denote

$$\tilde{J}_n^{(0,\alpha)} = 2J_n^{(0,\alpha)} - J_n^{(0,2\alpha)},$$

where, by definition,

$$J_n^{(0,\alpha)} = -\alpha \sum_{j=1}^n (I - \alpha \bar{A})^{n-j} \epsilon(Z_j).$$

Substituting and using the elementary identity  $X^m - Y^m = (X - Y) \sum_{i=1}^m X^{i-1} Y^{m-i}$  with  $X = I - \alpha \bar{A}$ ,  $Y = I - 2\alpha \bar{A}$ ,  $X - Y = \alpha \bar{A}$ , we get

$$\begin{aligned}
\tilde{J}_n^{(0,\alpha)} &= -2\alpha \sum_{j=1}^n \left[ (I - \alpha \bar{A})^{n-j} - (I - 2\alpha \bar{A})^{n-j} \right] \epsilon(Z_j) \\
&= -2\alpha^2 \bar{A} \sum_{j=1}^n \underbrace{\sum_{i=1}^{n-j} (I - \alpha \bar{A})^{i-1} (I - 2\alpha \bar{A})^{n-j-i}}_{=: H_j^{(n)}} \epsilon(Z_j) \\
&= -2\alpha^2 \bar{A} \sum_{j=1}^n H_j^{(n)} \epsilon(Z_j).
\end{aligned}$$

The extra factor  $\alpha \bar{A}$  pulled out front is the source of the additional  $\alpha$ -decay of the RR difference compared to a single LSA trajectory.

### 2.3. Norm Estimate for $H_j^{(n)}$

To bound  $\|H_j^{(n)}\|$  we use the following standard result.

**Lemma 1.** Let  $-\bar{A}$  be a Hurwitz matrix. Then for any  $P = P^\top \succ 0$  there exists a unique  $Q = Q^\top \succ 0$  satisfying

$$\bar{A}^\top Q + Q \bar{A} = P.$$

Moreover, letting

$$a := \frac{\lambda_{\min}(P)}{2 \|Q\|}, \quad \alpha_\infty := \min \left( \frac{\lambda_{\min}(P)}{2\kappa_Q \|\bar{A}\|_Q^2}, \frac{\|Q\|}{\lambda_{\min}(P)} \right),$$

with  $\kappa_Q := \lambda_{\max}(Q)/\lambda_{\min}(Q)$ , one has for all  $\alpha \in [0, \alpha_\infty]$ :

$$\alpha a \leq 1/2, \quad \|I - \alpha \bar{A}\|_Q^2 \leq 1 - \alpha a.$$

We now estimate  $\|H_j^{(n)}\|$  by combining the triangle inequality, submultiplicativity, the equivalence  $\|X\| \leq \kappa_Q^{1/2} \|X\|_Q$  (applied to **each** operator-norm factor, hence two factors of  $\kappa_Q^{1/2}$  multiplying to  $\kappa_Q$ ), and the Lyapunov contraction at step sizes  $\alpha$  and  $2\alpha$ :

$$\begin{aligned}
\|H_j^{(n)}\| &\leq \sum_{i=1}^{n-j} \|I - \alpha \bar{A}\|^{i-1} \|I - 2\alpha \bar{A}\|^{n-j-i} && \text{(triangle + submult.)} \\
&\leq \kappa_Q \sum_{i=1}^{n-j} (1 - \alpha a)^{(i-1)/2} (1 - 2\alpha a)^{(n-j-i)/2} && \text{(equiv. + Lyapunov)} \\
&= \kappa_Q (1 - \alpha a)^{(n-j-1)/2} \sum_{k=0}^{n-j-1} \left( \frac{1 - 2\alpha a}{1 - \alpha a} \right)^{k/2} && \text{(reindex } k = n - j - i) \\
&\leq \kappa_Q (1 - \alpha a)^{(n-j-1)/2} \frac{1}{1 - \sqrt{\frac{1-2\alpha a}{1-\alpha a}}} && \text{(geometric series).}
\end{aligned}$$

For  $\alpha a \leq 1/2$  the geometric rate is bounded below by an **elementary** inequality: combining  $\sqrt{(1-2\alpha a)/(1-\alpha a)} \leq \sqrt{1-\alpha a}$  (since  $(1-2\alpha a) \leq (1-\alpha a)^2$ ) with  $1 - \sqrt{1-x} \geq x/2$  on  $[0, 1]$  gives

$$1 - \sqrt{\frac{1-2\alpha a}{1-\alpha a}} \geq 1 - \sqrt{1-\alpha a} \geq \frac{\alpha a}{2}, \quad \text{i.e.} \quad \frac{1}{1 - \sqrt{\frac{1-2\alpha a}{1-\alpha a}}} \leq \frac{2}{\alpha a}.$$

Defining the chapter-local constant

$$\bar{C}_A := \kappa_Q$$

(distinct from the assumption-2 sup-norm constant  $C_A$ , which enters separately via  $\|\bar{A}\|$  in the next subsection), we arrive at the final estimate

$$\|H_j^{(n)}\| \leq \bar{C}_A (1 - \alpha a)^{(n-j-1)/2} \frac{2}{\alpha a}.$$

The kernel decays geometrically in  $n - j$  at rate  $\sqrt{1 - \alpha a}$  but its summed weight is of order  $1/(\alpha a)$ . The next subsection shows how the prefactor  $\alpha^2$  in  $\tilde{J}_n^{(0, \alpha)}$  exactly compensates this divergence.

## 2.4. $L^2$ Bound for the Zeroth-Order Term

The expression

$$\tilde{J}_n^{(0, \alpha)} = - \sum_{j=1}^n 2\alpha^2 \bar{A} H_j^{(n)} \epsilon(Z_j)$$

is a weighted sum of values of the centered noise function  $\epsilon$  along the Markov chain. We control its tail via the following Markov concentration statement.

**Lemma 2.** Assume **UGE 1**. Let  $\{g_i\}_{i=1}^n$  be a family of measurable functions  $g_i : Z \rightarrow \mathbb{R}^d$  such that

$$c_i = \|g_i\|_\infty < \infty \quad \text{for all } i \geq 1, \quad \pi(g_i) = 0 \quad \text{for all } i \in \{1, \dots, n\}.$$

Then, for any initial distribution  $\xi$  on  $(Z, \mathcal{Z})$ , any  $n \in \mathbb{N}$  and any  $t \geq 0$ ,

$$\mathbb{P}_\xi \left( \left\| \sum_{i=1}^n g_i(Z_i) \right\| \geq t \right) \leq 2 \exp \left( - \frac{t^2}{2u_n^2} \right),$$

where

$$u_n = 8 \left( \sum_{i=1}^n c_i^2 \right)^{1/2} \sqrt{t_{\text{mix}}}.$$

*Proof.* The proof follows the lines of Durmus et al. (2025, Lemma 9).

We apply the lemma with  $g_{j(z)} = -2\alpha^2 \bar{A} H_j^{(n)} \epsilon(z)$ . Each  $g_j$  is centered under  $\pi$  since  $\pi(\epsilon) = 0$ , so the centering hypothesis holds. Combining the bound  $\|\bar{A}\| \leq C_A$  (Assumption 2) with the previous subsection, define

$$\tilde{C}_A := C_A \bar{C}_A = C_A \kappa_Q.$$

Then the per-summand bound is

$$\begin{aligned} \|g_j\|_\infty &\leq 2\alpha^2 \tilde{C}_A (1 - \alpha a)^{(n-j-1)/2} \frac{2}{\alpha a} \|\epsilon\|_\infty \\ &= 4\alpha \tilde{C}_A \|\epsilon\|_\infty (1 - \alpha a)^{(n-j-1)/2}. \end{aligned}$$

Note the prefactor  $\alpha^2$  collapsing to  $\alpha$ : the  $1/(\alpha a)$  blow-up in  $H_j^{(n)}$  is exactly absorbed by one factor of  $\alpha$ . Squaring and summing over  $j$ , using  $1 - \alpha a \geq 1/2$ :

$$\begin{aligned} \sum_{j=1}^n \|g_j\|_\infty^2 &\leq 16\alpha^2 \tilde{C}_A^2 \|\epsilon\|_\infty^2 \sum_{j=1}^n (1 - \alpha a)^{n-j-1} \\ &\leq 16\alpha^2 \tilde{C}_A^2 \|\epsilon\|_\infty^2 \frac{1}{(1 - \alpha a)\alpha a} \\ &\leq \frac{32\alpha \tilde{C}_A^2 \|\epsilon\|_\infty^2}{a}. \end{aligned}$$

Plugging this into the variance proxy  $u_n^2 = 64 t_{\text{mix}} \sum_{j=1}^n \|g_j\|_\infty^2$  from the lemma and defining

$$\hat{C}_A := \sqrt{2048} \tilde{C}_A \|\epsilon\|_\infty \sqrt{\frac{t_{\text{mix}}}{a}} = 32\sqrt{2} \tilde{C}_A \|\epsilon\|_\infty \sqrt{\frac{t_{\text{mix}}}{a}},$$

gives

$$u_n^2 \leq \hat{C}_A^2 \alpha.$$

To convert the sub-Gaussian tail into a moment bound, we use the following standard fact.

**Lemma 3.** Let  $X$  be an  $\mathbb{R}^d$ -valued random variable satisfying

$$\mathbb{P}(\|X\| \geq t) \leq 2 \exp\left(-\frac{t^2}{2\sigma^2}\right) \quad \text{for all } t \geq 0,$$

for some  $\sigma^2 > 0$ . Then, for any  $p \geq 2$ , it holds that

$$\mathbb{E}[\|X\|^p] \leq 2 p^{p/2} \sigma^p.$$

Applying the lemma with  $\sigma^2 = u_n^2 \leq \hat{C}_A^2 \alpha$  and an arbitrary  $p \geq 2$  gives

$$\mathbb{E}^{1/p}[\|\tilde{J}_n^{(0,\alpha)}\|^p] \leq 2^{1/p} \sqrt{p} \hat{C}_A \sqrt{\alpha} \leq 2\sqrt{p} \hat{C}_A \sqrt{\alpha},$$

or equivalently

$$\frac{1}{\sqrt{\alpha}} \mathbb{E}^{1/p}[\|\tilde{J}_n^{(0,\alpha)}\|^p] \leq 2\sqrt{p} \hat{C}_A.$$

The zeroth-order RR difference is therefore  $O(\sqrt{\alpha})$  in  $L^p$  for every  $p \geq 2$ , uniformly in  $n$  — an order- $\sqrt{\alpha}$  gain over a single LSA trajectory, which is  $O(1)$  in  $L^p$  in the stationary regime. The  $p = 2$  statement used in the rest of the analysis is a special case.

### 3. Last Iterate Analysis

#### 3.1. Centered Bound for the Shifted First-Order Perturbation

The purpose of this section is to isolate the last-iteration weighted term which appears in the analysis of  $J_n^{(1,\alpha)}$ , and to record a clean  $L_p$  bound for its centered part. The proof uses the deterministic-product perturbation expansion from Samsonov et al. (2025, Proposition 9), specialized to the constant-stepsize setting.

Let

$$B = I - \alpha \bar{A}.$$

Throughout this section we assume  $0 < \alpha \leq \alpha_\infty$ , so that

$$\|B^m\|_Q \leq (1 - \alpha a)^{m/2}, \quad m \geq 0.$$

All constants denoted by  $C$  may depend on  $C_A$ ,  $\kappa_Q$ , and norm equivalence constants, but not on  $\alpha$ ,  $n$ ,  $p$ , or  $t_{\text{mix}}$ .

The deterministic-product components are defined by

$$J_n^{(0,\alpha)} = B J_{n-1}^{(0,\alpha)} - \alpha \epsilon(Z_n), \quad J_0^{(0,\alpha)} = 0,$$

and

$$J_n^{(1,\alpha)} = B J_{n-1}^{(1,\alpha)} - \alpha \tilde{A}(Z_n) J_{n-1}^{(0,\alpha)}, \quad J_0^{(1,\alpha)} = 0.$$

Hence

$$J_n^{(0,\alpha)} = -\alpha \sum_{k=1}^n B^{n-k} \epsilon(Z_k),$$

and

$$\begin{aligned} J_n^{(1,\alpha)} &= -\alpha \sum_{j=1}^n B^{n-j} \tilde{A}(Z_j) J_{j-1}^{(0,\alpha)} \\ &= \alpha^2 \sum_{1 \leq k < j \leq n} B^{n-j} \tilde{A}(Z_j) B^{j-1-k} \epsilon(Z_k) \\ &= \alpha^2 \sum_{k=1}^{n-1} \sum_{r=1}^{n-k} B^{n-k-r} \tilde{A}(Z_{k+r}) B^{r-1} \epsilon(Z_k). \end{aligned}$$

The sign in the second line is positive because  $J_{j-1}^{(0,\alpha)}$  already contains the minus sign.

The last display is for  $J_n^{(1,\alpha)}$  itself. The last-iteration estimate below operates on a **shifted** version, obtained by inserting one additional left factor  $B$  into the recursion:

$$S_n = \sum_{t=0}^{n-1} B^{n-t} \tilde{A}(Z_{t+1}) J_t^{(0,\alpha)}.$$

Reindexing  $j = t + 1$  rewrites  $S_n$  in the same form as  $J_n^{(1,\alpha)}$  but with one extra power of  $B$ :

$$S_n = \sum_{j=1}^n B^{n-j+1} \tilde{A}(Z_j) J_{j-1}^{(0,\alpha)} = -\frac{1}{\alpha} B J_n^{(1,\alpha)}.$$

Thus the corresponding shifted first-order contribution is



$$T_n^{(1,\alpha)} = -\alpha S_n = B J_n^{(1,\alpha)},$$

i.e.  $T_n^{(1,\alpha)}$  is exactly  $J_n^{(1,\alpha)}$  pre-multiplied by one additional  $B$ . In particular, any  $L_p$  bound on  $S_n$  (or  $T_n^{(1,\alpha)}$ ) implies the same bound on  $J_n^{(1,\alpha)}$  up to the contraction factor  $\|B\|_Q \leq 1$ .

**Lemma 4.** Assume the Markov chain is started from stationarity, that is, the law of  $Z_1$  is  $\pi$ , and  $\pi(\tilde{A}) = 0$ . For every  $p \geq 2$ ,

$$\|S_n - \mathbb{E}S_n\|_{L_p} \leq C \|\epsilon\|_\infty \left( p^{3/2} t_{\text{mix}}^{1/2} \frac{1}{a} + p^{1/2} t_{\text{mix}}^{3/2} \sqrt{\frac{\alpha}{a}} \right).$$

Consequently,

$$\|T_n^{(1,\alpha)} - \mathbb{E}T_n^{(1,\alpha)}\|_{L_p} \leq C\alpha \|\epsilon\|_\infty \left( p^{3/2} t_{\text{mix}}^{1/2} \frac{1}{a} + p^{1/2} t_{\text{mix}}^{3/2} \sqrt{\frac{\alpha}{a}} \right).$$

*Proof.* Since  $J_0^{(0,\alpha)} = 0$ , the term  $t = 0$  in  $S_n$  vanishes. Substituting the explicit formula for  $J_t^{(0,\alpha)}$  gives

$$\begin{aligned} S_n &= -\alpha \sum_{t=1}^{n-1} \sum_{k=1}^t B^{n-t} \tilde{A}(Z_{t+1}) B^{t-k} \epsilon(Z_k) \\ &= -\alpha \sum_{k=1}^{n-1} H_{k+1}^{(w)} \epsilon(Z_k), \end{aligned}$$

where, after the change of summation index  $l = t - k + 1$ ,

$$H_{k+1}^{(w)} = \sum_{l=1}^{n-k} B^{n-k-l+1} \tilde{A}(Z_{k+l}) B^{l-1}.$$

The kernel  $H_{k+1}^{(w)}$  acts on the past noise  $\epsilon(Z_k)$  through future states  $Z_{k+l}$ , so  $H_{k+1}^{(w)} \epsilon(Z_k)$  is a future-weighted bilinear functional of the trajectory.

Define

$$\mu_k^{(w)} = \mathbb{E}_\pi \left[ H_{k+1}^{(w)} \epsilon(Z_k) \right].$$

Then

$$S_n - \mathbb{E}S_n = -\alpha \sum_{k=1}^{n-1} \left\{ H_{k+1}^{(w)} \epsilon(Z_k) - \mu_k^{(w)} \right\}.$$

Let  $\mathcal{F}_k = \sigma(Z_1, \dots, Z_k)$ . The Markov property gives

$$\mathbb{E} \left[ H_{k+1}^{(w)} \epsilon(Z_k) \mid \mathcal{F}_k \right] = v_k^{(w,\epsilon)}(Z_k),$$

where

$$v_k^{(w)}(z) = \sum_{l=1}^{n-k} B^{n-k-l+1} (Q^l \tilde{A})(z) B^{l-1}, \quad v_k^{(w,\epsilon)}(z) = v_k^{(w)}(z) \epsilon(z).$$

Here  $Q$  denotes the one-step Markov transition kernel of  $(Z_k)_{k \geq 1}$ , acting on bounded matrix-valued functions by integration against the conditional law:

$$(Q\tilde{A})(z) = \int \tilde{A}(u) Q(z, du) = \mathbb{E}[\tilde{A}(Z_{k+1}) \mid Z_k = z],$$

and  $Q^l$  is its  $l$ -fold iterate, the  $l$ -step kernel  $Q^l(z, du) = \mathbb{P}(Z_{k+l} \in du \mid Z_k = z)$ , so that

$$(Q^l \tilde{A})(z) = \int \tilde{A}(u) Q^l(z, du) = \mathbb{E}[\tilde{A}(Z_{k+l}) \mid Z_k = z].$$

In particular  $(Q^l \tilde{A})(z) \rightarrow \pi(\tilde{A}) = 0$  at the geometric rate dictated by UGE, which is the only fact about  $Q^l$  used below.

Under stationarity,  $\mu_k^{(w)} = \pi(v_k^{(w, \varepsilon)})$ . Therefore

$$S_n - \mathbb{E}S_n = -\alpha(U_M + U_R),$$

with

$$U_M = \sum_{k=1}^{n-1} \left\{ H_{k+1}^{(w)} \epsilon(Z_k) - v_k^{(w, \varepsilon)}(Z_k) \right\}$$

and

$$U_R = \sum_{k=1}^{n-1} \left\{ v_k^{(w, \varepsilon)}(Z_k) - \pi(v_k^{(w, \varepsilon)}) \right\}.$$

This decomposition splits the centered statistic into two structurally different parts:  $U_R$  is an ordinary centered additive functional of the original Markov chain, while  $U_M$  is a future-centered bilinear term — the conditional expectation given  $\mathcal{F}_k$  vanishes summand-wise, but the summands are not forward martingale differences. We bound each piece separately.

*Step 1: bound on  $U_R$ .* Because  $\pi(\tilde{A}) = 0$ , applying  $Q^l$  followed by integration against  $\pi$  replaces  $\tilde{A}$  by its  $l$ -step propagation away from stationarity. UGE then gives the geometric Dobrushin bound

$$\|(Q^l \tilde{A})(z)\| = \left\| \int \tilde{A}(u) \{Q^l(z, du) - \pi(du)\} \right\| \leq 2C_A \Delta(Q^l) \leq 2C_A (1/4)^{\lfloor l/t_{\text{mix}} \rfloor}.$$

Inserting this into  $v_k^{(w)}$  and factoring out the slow rate  $(1 - \alpha a)^{(n-k)/2}$  from the two  $B$ -powers, the inner sum becomes a fast-decaying  $l$ -series:

$$\begin{aligned} \|v_k^{(w)}\|_\infty &\leq C \sum_{l=1}^{n-k} (1 - \alpha a)^{(n-k-l+1)/2} \Delta(Q^l) (1 - \alpha a)^{(l-1)/2} \quad (\text{triangle} + \text{Lyapunov}) \\ &\leq C (1 - \alpha a)^{(n-k)/2} \sum_{l=1}^{\infty} (1/4)^{\lfloor l/t_{\text{mix}} \rfloor} \quad (\text{extend to } l = \infty) \\ &\leq C t_{\text{mix}} (1 - \alpha a)^{(n-k)/2} \quad (\text{geometric block sum}). \end{aligned}$$

Multiplying by  $\|\epsilon\|_\infty$  gives

$$\|v_k^{(w, \varepsilon)}\|_\infty \leq C t_{\text{mix}} \|\epsilon\|_\infty (1 - \alpha a)^{(n-k)/2}.$$

The functions  $v_k^{(w, \varepsilon)}(Z_k) - \pi(v_k^{(w, \varepsilon)})$  are centered under  $\pi$  and uniformly bounded by the previous display, so the weighted Markov concentration/Rosenthal bound for centered time-dependent functions yields

$$\begin{aligned}
\|U_R\|_{L_p} &\leq Cp^{1/2}t_{\text{mix}}^{1/2} \left( \sum_{k=1}^{n-1} \|v_k^{(w,\varepsilon)}\|_{\infty}^2 \right)^{1/2} && \text{(Rosenthal)} \\
&\leq Cp^{1/2}t_{\text{mix}}^{3/2} \|\epsilon\|_{\infty} \left( \sum_{k=1}^{n-1} (1-\alpha a)^{n-k} \right)^{1/2} && \text{(plug previous bound)} \\
&\leq Cp^{1/2}t_{\text{mix}}^{3/2} \|\epsilon\|_{\infty} \frac{1}{\sqrt{\alpha a}} && \text{(geometric series).}
\end{aligned}$$

*Step 2: bound on  $U_M$ .* Project onto a deterministic unit vector  $u$  and unfold  $H_{k+1}^{(w)}u$  as a sum over future states:

$$H_{k+1}^{(w)}u = \sum_{l=1}^{n-k} g_{k,l}(Z_{k+l}), \quad g_{k,l}(z) = B^{n-k-l+1} \tilde{A}(z) B^{l-1} u.$$

Each  $g_{k,l}$  is centered under  $\pi$  (since  $\pi(\tilde{A}) = 0$ ), and the two  $B$ -powers give the uniform bound

$$\|g_{k,l}\|_{\infty} \leq C(1-\alpha a)^{(n-k)/2},$$

which is independent of  $l$ . Squaring and summing over  $l$  produces only linear growth in  $n-k$ :

$$\sum_{l=1}^{n-k} \|g_{k,l}\|_{\infty}^2 \leq C(n-k)(1-\alpha a)^{n-k}.$$

Note the contrast with the  $U_R$  analysis: there, UGE folded the  $l$ -sum into a single  $t_{\text{mix}}$  factor, whereas here the same  $l$ -independent bound is applied  $(n-k)$  times — the price of conditional centering being weaker than  $\pi$ -centering.

Applying the Markov concentration lemma to the future chain conditionally on  $\mathcal{F}_k$ ,

$$\|H_{k+1}^{(w)}u - \mathbb{E}[H_{k+1}^{(w)}u \mid \mathcal{F}_k]\|_{L_p} \leq Cp^{1/2}t_{\text{mix}}^{1/2} \sqrt{(n-k)(1-\alpha a)^{n-k}},$$

and multiplying by  $\epsilon(Z_k)$  (which is  $\mathcal{F}_k$ -measurable and bounded by  $\|\epsilon\|_{\infty}$ ) gives

$$\|H_{k+1}^{(w)}\epsilon(Z_k) - v_k^{(w,\varepsilon)}(Z_k)\|_{L_p} \leq Cp^{1/2}t_{\text{mix}}^{1/2} \|\epsilon\|_{\infty} \sqrt{(n-k)(1-\alpha a)^{n-k}}.$$

The summands in  $U_M$  are centered conditionally on  $\mathcal{F}_k$ , but they are *not* forward martingale differences: the kernel  $H_{k+1}^{(w)}$  peeks into the future of  $Z_k$ . To assemble the per- $k$  bounds into a bound on the sum we therefore invoke the weighted Burkholder/block-coupling estimate for future-centered bilinear Markov sums of Samsonov et al. (2025, Proposition 9), specialized to the present constant-stepsizes kernel:

$$\|U_M\|_{L_p} \leq Cp \left( \sum_{k=1}^{n-1} \|H_{k+1}^{(w)}\epsilon(Z_k) - v_k^{(w,\varepsilon)}(Z_k)\|_{L_p}^2 \right)^{1/2}.$$

Plugging in the previous bound and evaluating the resulting sum:

$$\begin{aligned}
\|U_M\|_{L_p} &\leq Cp^{3/2}t_{\text{mix}}^{1/2} \|\epsilon\|_{\infty} \left( \sum_{k=1}^{n-1} (n-k)(1-\alpha a)^{n-k} \right)^{1/2} && \text{(plug previous bound)} \\
&\leq Cp^{3/2}t_{\text{mix}}^{1/2} \|\epsilon\|_{\infty} \frac{1}{\alpha a} && \left( \text{use } \sum_m m x^m \leq Cx^{-2} \right).
\end{aligned}$$

The extra factor  $1/\sqrt{\alpha a}$  relative to  $U_R$  is precisely the cost of using conditional rather than stationary centering.

*Step 3: assembly.* Combining the two pieces via  $S_n - \mathbb{E}S_n = -\alpha(U_M + U_R)$  and the triangle inequality,

$$\begin{aligned} \|S_n - \mathbb{E}S_n\|_{L_p} &\leq \alpha (\|U_M\|_{L_p} + \|U_R\|_{L_p}) \\ &\leq C \|\epsilon\|_\infty \left( p^{3/2} t_{\text{mix}}^{1/2} \frac{1}{a} + p^{1/2} t_{\text{mix}}^{3/2} \sqrt{\frac{\alpha}{a}} \right). \end{aligned}$$

The first term (from  $U_M$ ) is the leading contribution for small  $\alpha$ ; the second (from  $U_R$ ) carries the heavier  $t_{\text{mix}}$  dependence but vanishes as  $\alpha \rightarrow 0$ . Multiplying by  $\alpha$  gives the asserted bound for  $T_n^{(1,\alpha)} = -\alpha S_n$ .  $\square$

**Remark 1.** The stationarity assumption is used only to avoid boundary notation in this local calculation. For a non-stationary initial law, UGE gives the same bulk estimate plus the standard exponentially small mixing boundary terms. The bias of  $T_n^{(1,\alpha)}$  is a separate issue and is controlled through the stationary expansion of  $\mathbb{E}J_n^{(1,\alpha)}$  in Levin et al. (2025).

### 3.2. Application to the PR-averaged RR Misadjustment

The PR-averaged Richardson–Romberg expansion produces, after Step (S8) of the Samsonov scheme applied separately at step sizes  $\alpha$  and  $2\alpha$ , a “misadjustment” remainder

$$D_1^{\text{mis, RR}} = \frac{1}{\sqrt{n}} \sum_{k=n_0}^{n-1} \left( 2J_k^{(1,\alpha)} - J_k^{(1,2\alpha)} \right),$$

whose centered part must be controlled to feed into a Berry–Esseen statement for  $\sqrt{n}(\bar{\theta}_n^{(\text{RR},\alpha)} - \theta^*)$ . The bias is not the obstruction: by Levin et al. (2025, Proposition 2),

$$\mathbb{E}_\pi [J_\infty^{(1,\alpha)}] = \alpha\Delta + O(\alpha^2),$$

so the linear term  $\alpha\Delta$  cancels in the RR-combination and the stationary bias of  $D_1^{\text{mis, RR}}$  is  $O(\alpha^2)$ . What remains is the centered fluctuation.

Define

$$\Phi(p, \alpha) := p^{3/2} \frac{t_{\text{mix}}^{1/2}}{a} + p^{1/2} t_{\text{mix}}^{3/2} \sqrt{\alpha/a}.$$

The lemma applied at  $\alpha$  and at  $2\alpha$  gives, separately,

$$\|T_n^{(1,\alpha)} - \mathbb{E}T_n^{(1,\alpha)}\|_{L_p} \leq C\alpha \Phi(p, \alpha), \quad \|T_n^{(1,2\alpha)} - \mathbb{E}T_n^{(1,2\alpha)}\|_{L_p} \leq C\alpha \Phi(p, 2\alpha) \leq C'\alpha \Phi(p, \alpha),$$

where  $C' = \sqrt{2}C$  absorbs the  $\sqrt{2}$ -factor coming from the  $2\alpha$  scaling. Combining the two by the triangle inequality and using the index-shift identity  $T_k^{(1,\alpha)} = B J_k^{(1,\alpha)}$  from the start of this section, we get

$$\left\| \left( 2J_k^{(1,\alpha)} - J_k^{(1,2\alpha)} \right) - \mathbb{E} \left( 2J_k^{(1,\alpha)} - J_k^{(1,2\alpha)} \right) \right\|_{L_p} \leq C\alpha \Phi(p, \alpha),$$

uniformly in  $k$ . PR-averaging through  $\frac{\sqrt{n}}{n-n_0}$  and absorbing the constant therefore yields

$$\|D_1^{\text{mis, RR}}\|_{L_p} = O(\sqrt{n}\alpha).$$

At the optimal scale  $\alpha \asymp n^{-1/2}$  this is  $O(1)$ , and as a remainder in a Berry–Esseen statement of order  $n^{-1/4}$  it is **useless**: the misadjustment must be at least  $o(1)$  to be subleading.

### 3.3. Richardson–Romberg Difference of $S_n$

The single- $\alpha$  centered bound, applied separately at  $\alpha$  and  $2\alpha$  and combined termwise via  $\|2X - Y\| \leq 2\|X\| + \|Y\|$ , produces a misadjustment estimate of order  $O(\sqrt{n}\alpha) = O(1)$  at  $\alpha \asymp n^{-1/2}$ . This estimate is **blind to coupling**: it never uses the fact that  $S_n^{(\alpha)}$  and  $S_n^{(2\alpha)}$  share the same noise realization  $\{Z_k\}$ ,

and is the one that holds for a generic centred difference of two centred quantities. The Richardson–Romberg construction nonetheless promises an additional factor of  $\alpha$  from the deterministic prefactors, via the elementary identity  $X^m - Y^m = (X - Y) \sum_{i=1}^m X^{i-1} Y^{m-i}$  at  $X = B_\alpha$ ,  $Y = B_{2\alpha}$ ,  $X - Y = \alpha \bar{A}$ .

We accordingly view the Richardson–Romberg difference of  $S_n^{(\alpha)}$  as a single weighted sum on the common noise, with a deterministic RR-kernel in place of the single- $\alpha$  kernel  $H^{(w)}$ . The structure of the analysis matches the previous subsection — single-kernel form, conditional/stationary centering, Markov-Rosenthal and weighted Burkholder — with one new ingredient, a pointwise bound on the RR-kernel that uses the identity above to extract a factor of  $\alpha$  uniformly across the kernel.

### 3.3.1. Single-Kernel Form

For step sizes  $\alpha$  and  $2\alpha$ , write  $B_\alpha = I - \alpha \bar{A}$ ,  $B_{2\alpha} = I - 2\alpha \bar{A}$ , and consider the **Richardson–Romberg difference** of the shifted statistic at fixed horizon  $n$ :

$$\tilde{S}_n^{\text{RR}} := 2 S_n^{(\alpha)} - S_n^{(2\alpha)}.$$

Both  $S_n^{(\alpha)}$  and  $S_n^{(2\alpha)}$  are driven by the **same** noise realization  $\{Z_k\}$ . Substituting  $S_n^{(\alpha)} = -\alpha \sum_{k=1}^{n-1} H_{k+1}^{(w),\alpha} \epsilon(Z_k)$  from the previous subsection at both step sizes and combining,

$$\tilde{S}_n^{\text{RR}} = -2\alpha \sum_{k=1}^{n-1} \left[ H_{k+1}^{(w),\alpha} - H_{k+1}^{(w),2\alpha} \right] \epsilon(Z_k) = -\sum_{k=1}^{n-1} \mathcal{H}_{k+1}^{\text{RR}} \epsilon(Z_k),$$

with the **RR kernel**

$$\mathcal{H}_{k+1}^{\text{RR}} := 2\alpha \sum_{l=1}^{n-k} \left[ B_\alpha^{a_l} \tilde{A}(Z_{k+l}) B_\alpha^{b_l} - B_{2\alpha}^{a_l} \tilde{A}(Z_{k+l}) B_{2\alpha}^{b_l} \right], \quad a_l := n - k - l + 1, \quad b_l := l - 1.$$

By construction  $a_l + b_l = n - k$ . The future-state factor  $\tilde{A}(Z_{k+l})$  enters  $\mathcal{H}^{\text{RR}}$  in exactly the same way as in  $H^{(w)}$ : only the deterministic prefactors  $B_\alpha^{a_l}, B_{2\alpha}^{a_l}$  carry the RR cancellation.

### 3.3.2. Pointwise Kernel Bound

**Lemma 5.** Let  $\alpha, 2\alpha \in (0, \alpha_\infty]$ , set  $\bar{C}_{\text{RR}} := \kappa_Q^{3/2} \|\bar{A}\|$ , and let  $M \in \mathbb{R}^{d \times d}$  be any deterministic matrix. For all integers  $j, m \geq 0$  with  $j + m \geq 1$ ,

$$\| B_\alpha^j M B_\alpha^m - B_{2\alpha}^j M B_{2\alpha}^m \| \leq \bar{C}_{\text{RR}} \alpha (j + m) (1 - \alpha a)^{(j+m-1)/2} \| M \|.$$

*Proof.* Apply the additive split

$$B_\alpha^j M B_\alpha^m - B_{2\alpha}^j M B_{2\alpha}^m = (B_\alpha^j - B_{2\alpha}^j) M B_\alpha^m + B_{2\alpha}^j M (B_\alpha^m - B_{2\alpha}^m),$$

and the elementary telescope  $X^r - Y^r = \sum_{i=1}^r X^{i-1} (X - Y) Y^{r-i}$  with  $X = B_\alpha$ ,  $Y = B_{2\alpha}$ ,  $X - Y = \alpha \bar{A}$ . For the first piece, submultiplicativity in  $\|\cdot\|_Q$  followed by Lyapunov contraction at both step sizes (with  $1 - 2\alpha a \leq 1 - \alpha a$ ) gives

$$\| B_\alpha^j - B_{2\alpha}^j \|_Q \leq \alpha \|\bar{A}\|_Q j (1 - \alpha a)^{(j-1)/2},$$

hence after one application of the equivalence  $\|\cdot\| \leq \kappa_Q^{1/2} \|\cdot\|_Q$ ,

$$\| (B_\alpha^j - B_{2\alpha}^j) M B_\alpha^m \| \leq \alpha \kappa_Q^{3/2} \|\bar{A}\| j (1 - \alpha a)^{(j+m-1)/2} \| M \|.$$

The second piece is symmetric and gives the same bound with  $j$  replaced by  $m$ . Adding the two yields the claim.  $\square$

The lemma is the **only** place where Richardson–Romberg cancellation is used. Two structural features deserve emphasis:

- The factor  $\alpha$  extracted by the telescope is the deterministic manifestation of RR coupling: it is the same identity that powered the zeroth-order bound on  $\tilde{J}_n^{(0,\alpha)}$  in Chapter 2 and the discrete-derivative bound on  $\mathcal{Q}_{l+1}^{\text{RR}} - \mathcal{Q}_l^{\text{RR}}$  in Chapter 4.
- The factor  $j + m$  is the count of telescope summands. It cannot be removed without strengthening the geometric decay, which is already optimal for  $B_\alpha$ . In the present application  $j + m = n - k$  is the **full lag**, and this growth is what spoils the gain in the variance proxies below.

### 3.3.3. Centered Decomposition

Following the single- $\alpha$  scheme, fix  $\mathcal{F}_k = \sigma(Z_1, \dots, Z_k)$ . Under stationarity ( $Z_1$  drawn from  $\pi$ ,  $\pi(\tilde{A}) = 0$ ), the Markov property gives

$$\mathbb{E}[\mathcal{H}_{k+1}^{\text{RR}} \epsilon(Z_k) | \mathcal{F}_k] = v_k^{(w, \text{RR}, \epsilon)}(Z_k),$$

with

$$v_k^{(w, \text{RR})}(z) := 2\alpha \sum_{l=1}^{n-k} [B_\alpha^{a_l}(Q^l \tilde{A})(z) B_\alpha^{b_l} - B_{2\alpha}^{a_l}(Q^l \tilde{A})(z) B_{2\alpha}^{b_l}], \quad v_k^{(w, \text{RR}, \epsilon)}(z) := v_k^{(w, \text{RR})}(z) \epsilon(z),$$

and  $\mathbb{E}[v_k^{(w, \text{RR}, \epsilon)}(Z_k)] = \pi(v_k^{(w, \text{RR}, \epsilon)})$ . Therefore

$$\tilde{S}_n^{\text{RR}} - \mathbb{E}\tilde{S}_n^{\text{RR}} = -(U_M^{\text{RR}} + U_R^{\text{RR}}),$$

$$U_M^{\text{RR}} = \sum_{k=1}^{n-1} \left\{ \mathcal{H}_{k+1}^{\text{RR}} \epsilon(Z_k) - v_k^{(w, \text{RR}, \epsilon)}(Z_k) \right\}, \quad U_R^{\text{RR}} = \sum_{k=1}^{n-1} \left\{ v_k^{(w, \text{RR}, \epsilon)}(Z_k) - \pi(v_k^{(w, \text{RR}, \epsilon)}) \right\}.$$

The decomposition has the same conditional/stationary split as in the single- $\alpha$  proof; the RR-kernel bound feeds into the variance proxies of each piece.

### 3.3.4. $L_p$ Bound

**Lemma 6.** Assume the Markov chain is started from stationarity and  $\pi(\tilde{A}) = 0$ . For every  $p \geq 2$ ,

$$\|U_R^{\text{RR}}\|_{L_p} \leq C \|\epsilon\|_\infty p^{1/2} t_{\text{mix}}^{3/2} \sqrt{\alpha}/a^{3/2},$$

$$\|U_M^{\text{RR}}\|_{L_p} \leq C \|\epsilon\|_\infty p^{3/2} t_{\text{mix}}^{1/2}/a^2,$$

with a constant  $C$  depending only on  $C_A, \kappa_Q, \|\bar{A}\|$ . Consequently,

$$\|\tilde{S}_n^{\text{RR}} - \mathbb{E}\tilde{S}_n^{\text{RR}}\|_{L_p} \leq C \|\epsilon\|_\infty \left( p^{3/2} t_{\text{mix}}^{1/2}/a^2 + p^{1/2} t_{\text{mix}}^{3/2} \sqrt{\alpha}/a^{3/2} \right).$$

*Proof.* The structure mirrors the single- $\alpha$  proof; only the variance proxies change. We trace each piece.

**Step 1 ( $U_R^{\text{RR}}$ ).** By UGE applied to the inner  $Q^l \tilde{A}$  factor,  $\|(Q^l \tilde{A})(z)\| \leq 2C_A(1/4)^{\lfloor l/t_{\text{mix}} \rfloor}$ . Combining with the kernel-difference Lemma (with  $j + m = n - k$  uniform in  $l$ , and  $\|(Q^l \tilde{A})(z)\|$  playing the role of  $\|M\|$ ),

$$\begin{aligned} \|v_k^{(w, \text{RR})}(z)\| &\leq 2\alpha \sum_{l=1}^{n-k} \bar{C}_{\text{RR}} \alpha (n-k) (1-\alpha a)^{(n-k-1)/2} 2C_A (1/4)^{\lfloor l/t_{\text{mix}} \rfloor} (\text{kernel} + \text{UGE}) \\ &\leq C t_{\text{mix}} \alpha^2 (n-k) (1-\alpha a)^{(n-k-1)/2} \quad (\text{geometric block sum}). \end{aligned}$$

Multiplying by  $\|\epsilon\|_\infty$ ,

$$\|v_k^{(w, \text{RR}, \epsilon)}\|_\infty \leq C t_{\text{mix}} \|\epsilon\|_\infty \alpha^2 (n-k) (1-\alpha a)^{(n-k-1)/2}.$$

Squaring and summing over  $k$ , with  $\sum_{m \geq 1} m^2 (1-\alpha a)^{m-1} \leq 2/(\alpha a)^3$ ,

$$\sum_{k=1}^{n-1} \|v_k^{(w, \text{RR}, \epsilon)}\|_\infty^2 \leq C^2 t_{\text{mix}}^2 \|\epsilon\|_\infty^2 \alpha^4 \frac{2}{(\alpha a)^3} = C' t_{\text{mix}}^2 \|\epsilon\|_\infty^2 \alpha / a^3.$$

The weighted Markov concentration of Levin et al. (2025, Lemma 11) applied to the centered functions  $v_k^{(w, \text{RR}, \epsilon)}(Z_k) - \pi(v_k^{(w, \text{RR}, \epsilon)})$  then gives

$$\|U_M^{\text{RR}}\|_{L_p} \leq C p^{1/2} t_{\text{mix}}^{1/2} \left( \sum_{k=1}^{n-1} \|v_k^{(w, \text{RR}, \epsilon)}\|_\infty^2 \right)^{1/2} \leq C p^{1/2} t_{\text{mix}}^{3/2} \|\epsilon\|_\infty \sqrt{\alpha} / a^{3/2}.$$

**Step 2 ( $U_M^{\text{RR}}$ ).** Project onto a unit vector  $u$  and unfold the kernel as a future-state sum,

$$\mathcal{H}_{k+1}^{\text{RR}} u = \sum_{l=1}^{n-k} F_l^{\text{RR}}(Z_{k+l}), \quad F_l^{\text{RR}}(z) := 2\alpha [B_\alpha^{a_l} \tilde{A}(z) B_\alpha^{b_l} - B_{2\alpha}^{a_l} \tilde{A}(z) B_{2\alpha}^{b_l}] u.$$

Each  $F_l^{\text{RR}}$  is centered under  $\pi$  and the kernel-difference Lemma gives the  **$l$ -uniform** bound

$$\|F_l^{\text{RR}}\|_\infty \leq 2\alpha \bar{C}_{\text{RR}} \alpha (a_l + b_l) (1 - \alpha a)^{(n-k-1)/2} \|\tilde{A}\| \leq C \alpha^2 (n-k) (1 - \alpha a)^{(n-k-1)/2}.$$

Squaring and summing over the  $n-k$  values of  $l$ ,

$$\sum_{l=1}^{n-k} \|F_l^{\text{RR}}\|_\infty^2 \leq C^2 \alpha^4 (n-k)^3 (1 - \alpha a)^{n-k-1}.$$

Markov concentration on the future chain conditionally on  $\mathcal{F}_k$  then yields

$$\|\mathcal{H}_{k+1}^{\text{RR}} u - \mathbb{E}[\mathcal{H}_{k+1}^{\text{RR}} u | \mathcal{F}_k]\|_{L_p, Z_k} \leq C p^{1/2} t_{\text{mix}}^{1/2} \alpha^2 (n-k)^{3/2} (1 - \alpha a)^{(n-k-1)/2},$$

and multiplying by  $\|\epsilon(Z_k)\| \leq \|\epsilon\|_\infty$  preserves the bound. Summing the squares over  $k$ , with  $\sum_{m \geq 1} m^3 (1 - \alpha a)^{m-1} \leq 6/(\alpha a)^4$ ,

$$\sum_{k=1}^{n-1} \|\mathcal{H}_{k+1}^{\text{RR}} \epsilon(Z_k) - v_k^{(w, \text{RR}, \epsilon)}(Z_k)\|_{L_p}^2 \leq C p t_{\text{mix}} \|\epsilon\|_\infty^2 \alpha^4 / (\alpha a)^4 = C p t_{\text{mix}} \|\epsilon\|_\infty^2 / a^4.$$

The weighted Burkholder estimate for future-centered bilinear Markov sums (Samsonov et al., 2025, Proposition 9; same instance as in the single- $\alpha$  proof) closes the bound

$$\|U_M^{\text{RR}}\|_{L_p} \leq C p \left( \sum_{k=1}^{n-1} \|\mathcal{H}_{k+1}^{\text{RR}} \epsilon(Z_k) - v_k^{(w, \text{RR}, \epsilon)}(Z_k)\|_{L_p}^2 \right)^{1/2} \leq C p^{3/2} t_{\text{mix}}^{1/2} \|\epsilon\|_\infty / a^2.$$

**Assembly.** Adding the two bounds gives the claim.  $\square$

### 3.3.5. Two Approaches: Triangle Inequality vs. Kernel Difference

There are two natural ways to bound the centered RR-difference, and it is instructive to lay them side by side.

**Approach A: triangle inequality on the single- $\alpha$  lemma.** The single- $\alpha$  centered bound from the previous subsection,

$$\|S_n^{(\alpha)} - \mathbb{E} S_n^{(\alpha)}\|_{L_p} \leq C \|\epsilon\|_\infty \left( p^{3/2} t_{\text{mix}}^{1/2} / a + p^{1/2} t_{\text{mix}}^{3/2} \sqrt{\alpha / a} \right),$$

applied separately at  $\alpha$  and  $2\alpha$  and combined by  $\|2X - Y\| \leq 2\|X\| + \|Y\|$ , yields

$$\|\tilde{S}_n^{\text{RR}} - \mathbb{E} \tilde{S}_n^{\text{RR}}\|_{L_p} \leq C \|\epsilon\|_\infty \left( p^{3/2} t_{\text{mix}}^{1/2} / a + p^{1/2} t_{\text{mix}}^{3/2} \sqrt{\alpha / a} \right).$$

This bound is **blind to coupling**: the same-noise structure of  $S_n^{(\alpha)}$  and  $S_n^{(2\alpha)}$  does not enter, and the resulting estimate is the one that holds for a generic centred difference of two centred statistics.

**Approach B: kernel-difference identity.** Express  $\tilde{S}_n^{\text{RR}}$  as a single noise-weighted sum with kernel  $\mathcal{H}^{\text{RR}}$  and feed the kernel-difference Lemma through the conditional/stationary centring scheme. The resulting bound, derived above, is

$$\| \tilde{S}_n^{\text{RR}} - \mathbb{E} \tilde{S}_n^{\text{RR}} \|_{L_p} \leq C \| \epsilon \|_{\infty} \left( p^{3/2} t_{\text{mix}}^{1/2} / a^2 + p^{1/2} t_{\text{mix}}^{3/2} \sqrt{\alpha} / a^{3/2} \right),$$

i.e. the same dependence on  $\alpha$  but an **additional** factor of  $1/a$  in both terms. For  $a < 1$  — the regime of interest — Approach B is **strictly worse** than Approach A.

The mechanism is structural. The kernel-difference identity extracts a factor of  $\alpha$  on each summand of  $\mathcal{H}^{\text{RR}}$ , but at the price of a count  $a_l + b_l = n - k$  of summands carrying that gain **uniformly in  $l$** . Squaring and summing over  $l$  in the Markov-Rosenthal/Burkholder variance proxy, the  $(n - k)$  factor combines with  $\sum (n - k)^q (1 - \alpha a)^{n-k-1} \leq q! / (\alpha a)^{q+1}$  to introduce one extra  $1/(\alpha a)$ , which is not absorbed by the  $\alpha^2$  kernel prefactor.

The same phenomenon, with weaker symptoms, was already visible in the zeroth-order analysis of Chapter 2: the bound  $\| H_j^{(n)} \| \leq \bar{C}_A (1 - \alpha a)^{(n-j-1)/2} 2/(\alpha a)$  has the same telescope-times-geometric-tail structure, and the  $1/(\alpha a)$  blow-up there is absorbed by **one** of the two  $\alpha$ -factors of the prefactor  $\alpha^2$  — leaving a residual  $1/a$  in the variance proxy.

In short, the kernel-difference identity captures only the **deterministic** part of the RR cancellation, through  $B_{\alpha}^m - B_{2\alpha}^m$ . It does not see the stochastic coupling through  $\{Z_k\}$ , and the latter is what the RR construction is for. The kernel-difference route therefore replaces a coupling-blind triangle estimate by an equally coupling-blind kernel estimate, with no net gain on the centered  $L_p$  norm of  $\tilde{S}_n^{\text{RR}}$ .

### 3.3.6. Implication for the PR-Averaged Misadjustment

Both approaches feed into the misadjustment in the same way. After  $\sqrt{n}$ -scaled PR-averaging,

$$\| D_1^{\text{mis, RR}} \|_{L_p} = O(\sqrt{n} \alpha) = O(1) \quad \text{at} \quad \alpha \asymp n^{-1/2},$$

which fails to vanish and does not match the martingale Berry–Esseen scale  $n^{-1/4}$ . The RR construction provides **no improvement** at this level of analysis: the rate is the same as for the single- $\alpha$  misadjustment.

A bound of order  $n^{-1/4}$  on the misadjustment is nevertheless reachable, but through a **different** decomposition. The depth-two perturbation expansion

$$H^{(0,\alpha)} = J^{(1,\alpha)} + J^{(2,\alpha)} + H^{(2,\alpha)}$$

of Levin et al. (2025) splits the misadjustment into three independently controlled pieces:

- the centered bilinear statistic  $\sum_t \tilde{A}(Z_{t+1}) J_t^{(0,\alpha)}$  is bounded by Levin’s Proposition 3 / Corollary 6 at order  $\sqrt{\alpha r} + 1/\sqrt{\alpha}$  in  $L_p$ , contributing  $O(n^{-1/4})$  after  $\sqrt{n}$ -scaled PR-averaging;
- the high-order moments  $\| J_n^{(2,\alpha)} \|_{L_p}$  and  $\| H_n^{(2,\alpha)} \|_{L_p}$  are  $O(\alpha^{3/2})$  by Levin’s Propositions 8–9, giving the same  $O(n^{-1/4})$  contribution after PR-averaging and  $\sqrt{n}$ -scaling;
- the bias of  $J^{(1,\alpha)}$  has the form  $\alpha \Delta + O(\alpha^2)$  by Levin’s Proposition 2, and the leading  $\alpha \Delta$  cancels under RR, leaving a stationary bias of order  $\alpha^2 = O(n^{-1})$ , well below the target.

Stitching these pieces together yields  $\| D_1^{\text{mis, RR}} \|_{L_p} = O(n^{-1/4})$ . The kernel-difference Lemma proved earlier in this section is **not** a load-bearing ingredient for that rate.

A **strict** improvement of the misadjustment to  $O(n^{-1/2})$ , subleading to martingale Berry–Esseen, would require a tool that exploits the stochastic coupling of the two trajectories — for instance, Berbee-style



decoupling on the joint two-chain state space, or a depth-two RR cancellation directly on  $J^{(2)}$ . We do not pursue this direction here.

## 4. Richardson–Romberg PR Weight Bounds

### 4.1. PR-Averaged Error Decomposition and the RR Weight

The starting point for all subsequent estimates is the **explicit** representation of  $\sqrt{n}(\bar{\theta}_n^{(\text{RR}, \alpha)} - \theta^*)$  as a noise-weighted sum with a deterministic kernel. This subsection derives that representation step by step.

**Depth-one expansion.** Unfolding the error recursion of Chapter 1 gives, for every  $k \geq 1$ ,

$$\theta_k^{(\alpha)} - \theta^* = -\alpha \sum_{l=1}^k \Gamma_{l+1:k}^{(\alpha)} \epsilon(Z_l) + \Gamma_{1:k}^{(\alpha)} (\theta_0 - \theta^*), \quad \Gamma_{l+1:k}^{(\alpha)} := \prod_{j=l+1}^k (I - \alpha A(Z_j)).$$

Replacing each random product  $\Gamma_{l+1:k}^{(\alpha)}$  by its deterministic counterpart  $B_\alpha^{k-l} := (I - \alpha \bar{A})^{k-l}$  and absorbing the difference into a higher-order remainder yields the **depth-one decomposition** (Samsonov et al., 2025, Proposition 9):

$$\theta_k^{(\alpha)} - \theta^* = -\alpha \sum_{l=1}^k B_\alpha^{k-l} \epsilon(Z_l) + B_\alpha^k (\theta_0 - \theta^*) + R_k^{(\alpha)},$$

where  $R_k^{(\alpha)} := J_k^{(1, \alpha)} + H_k^{(1, \alpha)}$  collects all contributions of order  $\alpha^{3/2}$  or higher in  $L_p$ . The first sum is the **depth-zero** term and is the only piece that carries the limiting Gaussian.

**PR averaging produces  $Q_l^{(\alpha)}$ .** Recall the PR average  $\bar{\theta}_n^{(\alpha)} = (n - n_0)^{-1} \sum_{k=n_0}^{n-1} \theta_k^{(\alpha)}$ . For notational clarity we set  $n_0 = 0$  from this point on (the burn-in adds only an exponentially small transient). Subtracting  $\theta^*$ , substituting the depth-one decomposition above, and **exchanging the order of summation** in the depth-zero piece,

$$\sum_{k=0}^{n-1} \sum_{l=1}^k B_\alpha^{k-l} \epsilon(Z_l) = \sum_{l=1}^{n-1} \left( \sum_{k=l}^{n-1} B_\alpha^{k-l} \right) \epsilon(Z_l),$$

isolates each noise sample  $\epsilon(Z_l)$  together with the deterministic **PR weight**

$$Q_l^{(\alpha)} := \alpha \sum_{k=l}^{n-1} B_\alpha^{k-l} = \alpha \sum_{j=0}^{n-l-1} B_\alpha^j.$$

Operationally,  $Q_l^{(\alpha)}$  collects the cumulative contribution of  $\epsilon(Z_l)$  to all averaged iterates  $\theta_l, \theta_{l+1}, \dots, \theta_{n-1}$ . Multiplying the resulting identity by  $\sqrt{n}$  gives the PR-averaged decomposition

$$\sqrt{n}(\bar{\theta}_n^{(\alpha)} - \theta^*) = W^{(\alpha)} + D^{(\alpha)},$$

where the **leading martingale-like sum** is

$$W^{(\alpha)} := -\frac{1}{\sqrt{n}} \sum_{l=1}^{n-1} Q_l^{(\alpha)} \epsilon(Z_l),$$

and the remainder  $D^{(\alpha)}$  packs the deterministic transient  $D_{\text{tr}}^{(\alpha)}$  and the higher-order stochastic part  $D_R^{(\alpha)}$ :

$$D^{(\alpha)} := D_{\text{tr}}^{(\alpha)} + D_R^{(\alpha)}, \quad D_{\text{tr}}^{(\alpha)} := \frac{1}{\sqrt{n}} \sum_{k=0}^{n-1} B_\alpha^k (\theta_0 - \theta^*), \quad D_R^{(\alpha)} := \frac{1}{\sqrt{n}} \sum_{k=0}^{n-1} R_k^{(\alpha)}.$$

The first sum is a deterministic geometric tail and becomes exponentially small after a logarithmic burn-in  $n_0 \asymp \log(n)/(a\alpha)$  (Lyapunov contraction  $\|B_\alpha^{n_0}\|_Q \leq e^{-1}$ ). The second is the source of the leading non-Gaussian correction in the Berry–Esseen rate; its RR-combination  $2D_R^{(\alpha)} - D_R^{(2\alpha)}$  is exactly the **misadjustment**  $D_1^{\text{mis, RR}}$  controlled in Chapters 2 and 3.

**RR combination produces  $\mathcal{Q}_l^{\text{RR}}$ .** The Richardson–Romberg iterate  $\bar{\theta}_n^{(\text{RR}, \alpha)} := 2\bar{\theta}_n^{(\alpha)} - \bar{\theta}_n^{(2\alpha)}$  inherits the PR decomposition **by linearity**: applying the previous display at step sizes  $\alpha$  and  $2\alpha$  separately and combining,

$$\sqrt{n} \left( \bar{\theta}_n^{(\text{RR}, \alpha)} - \theta^* \right) = W^{\text{RR}} + D^{\text{RR}},$$

$$W^{\text{RR}} := 2W^{(\alpha)} - W^{(2\alpha)} = -\frac{1}{\sqrt{n}} \sum_{l=1}^{n-1} \mathcal{Q}_l^{\text{RR}} \epsilon(Z_l),$$

where the **RR weight** is

$$\mathcal{Q}_l^{\text{RR}} := 2Q_l^{(\alpha)} - Q_l^{(2\alpha)}.$$

Crucially, both PR averages share the **same** noise realization  $\{Z_k\}$ , so the bracket  $\mathcal{Q}_l^{\text{RR}}$  is a single deterministic matrix kernel: all the stochastic content of  $W^{\text{RR}}$  now lives in  $\epsilon(Z_l)$  alone.

**Why  $\mathcal{Q}_l^{\text{RR}}$  is the right object.** Two questions about  $W^{\text{RR}}$  drive the rest of the chapter, and both answer themselves in terms of  $\mathcal{Q}_l^{\text{RR}}$ :

1. **Variance comparison.** The CLT identifies the limiting covariance of  $W^{\text{RR}}$  as  $\Sigma_\infty = \bar{A}^{-1} \Sigma_\epsilon^{(\text{M})} \bar{A}^{-\top}$  (the Markov-chain CLT covariance), which is exactly what one obtains if every  $\mathcal{Q}_l^{\text{RR}}$  is replaced by the asymptotic weight  $\bar{A}^{-1}$ . The finite- $n$  deviation is therefore controlled by  $\sum_l \|\mathcal{Q}_l^{\text{RR}} - \bar{A}^{-1}\|^2$  — see Section 4.5.
2. **Poisson-equation / Abel-summation remainder.** The standard Berry–Esseen route for Markov-chain noise solves the Poisson equation  $\hat{\epsilon} - Q\hat{\epsilon} = \epsilon$ , replaces  $\epsilon(Z_l)$  by  $\hat{\epsilon}(Z_l) - \hat{\epsilon}(Z_{l+1})$  (up to a martingale increment), and Abel-sums against the weight sequence  $\mathcal{Q}_l^{\text{RR}}$ . The resulting remainder has norm bounded by the **total variation**  $\sum_l \|\mathcal{Q}_{l+1}^{\text{RR}} - \mathcal{Q}_l^{\text{RR}}\|$ .

Both quantities are estimated in Sections 4.3–4.4 below. The closed-form identities of Section 4.2 reduce them to elementary operator-norm calculations on  $B_\alpha^m - B_{2\alpha}^m$ .

## 4.2. Closed-Form Identities

Throughout this chapter write

$$B_\alpha := I - \alpha \bar{A}, \quad B_{2\alpha} := I - 2\alpha \bar{A}, \quad k := n - l.$$

We assume  $\alpha, 2\alpha \in (0, \alpha_\infty]$ , so that the Lyapunov contraction  $\|B_\alpha^m\|_Q^2 \leq (1 - \alpha a)^m$  and  $\|B_{2\alpha}^m\|_Q^2 \leq (1 - 2\alpha a)^m$  hold (and we use freely  $1 - 2\alpha a \leq 1 - \alpha a$ ).

The geometric-series identity  $\sum_{j=0}^{m-1} B_\alpha^j = (\alpha \bar{A})^{-1} (I - B_\alpha^m)$  converts into the closed form

$$Q_l^{(\alpha)} = \alpha \sum_{j=0}^{n-l-1} B_\alpha^j = \alpha (\alpha \bar{A})^{-1} (I - B_\alpha^{n-l}) = \bar{A}^{-1} (I - B_\alpha^{n-l}).$$

This makes the convergence  $Q_l^{(\alpha)} \rightarrow \bar{A}^{-1}$  as  $k = n - l \rightarrow \infty$  explicit: the only obstruction is the geometric Lyapunov tail  $B_\alpha^k$ . Two immediate consequences are

$$Q_l^{(\alpha)} - \bar{A}^{-1} = -\bar{A}^{-1} B_\alpha^{n-l}, \quad Q_{l+1}^{(\alpha)} - Q_l^{(\alpha)} = -\alpha B_\alpha^{n-l-1}.$$

The first is the **asymptotic-weight error** and decays geometrically in  $k$ . The second is the **discrete derivative** used in Abel summation; note the explicit factor  $\alpha$  in front.

Applying both identities at  $\alpha$  and  $2\alpha$  and combining yields the basic **RR identities**:

$$\mathcal{Q}_l^{\text{RR}} - \bar{A}^{-1} = -\bar{A}^{-1} (2B_\alpha^k - B_{2\alpha}^k),$$

$$\mathcal{Q}_{l+1}^{\text{RR}} - \mathcal{Q}_l^{\text{RR}} = -2\alpha (B_\alpha^{k-1} - B_{2\alpha}^{k-1}).$$

These are the exact expressions bounded in the next subsection. Note the structural asymmetry between them:

- The asymptotic-weight error  $2B_\alpha^k - B_{2\alpha}^k$  is at the **single-trajectory** rate  $(1 - \alpha a)^{k/2}$  — the triangle inequality  $|2X - Y| \leq 2|X| + |Y|$  does not see the RR coupling, and indeed the leading  $-\bar{A}^{-1}$  in  $\mathcal{Q}_l^{\text{RR}}$  is the same as in the single-step weight  $Q_l^{(\alpha)}$ .
- The discrete derivative  $B_\alpha^{k-1} - B_{2\alpha}^{k-1}$ , however, is a **true** difference of contractions evaluated at the same exponent. The elementary identity  $X^m - Y^m = (X - Y) \sum_{i=1}^m X^{i-1} Y^{m-i}$  with  $X = B_\alpha$ ,  $Y = B_{2\alpha}$ ,  $X - Y = \alpha \bar{A}$  extracts an extra factor of  $\alpha$ , gaining one full power of  $\alpha$  over the single-step case  $Q_{l+1}^{(\alpha)} - Q_l^{(\alpha)} = -\alpha B_\alpha^{k-1}$ .

This local  $\alpha$ -gain in the discrete derivative is the **only** manifestation of Richardson–Romberg cancellation visible at the level of the PR weights.

### 4.3. Pointwise Bounds for the RR Weights

**Lemma 7.** Let  $\alpha, 2\alpha \in (0, \alpha_\infty]$ , set  $C_Q := \kappa_Q^{1/2} \|\bar{A}^{-1}\|$  and  $\tilde{C}_A := \kappa_Q \|\bar{A}\|$ , and write  $k = n - l$ .

(i) For every  $1 \leq l \leq n - 1$ ,

$$\|\mathcal{Q}_l^{\text{RR}} - \bar{A}^{-1}\| \leq 3C_Q(1 - \alpha a)^{k/2}.$$

(ii) For every  $1 \leq l \leq n - 2$ ,

$$\|\mathcal{Q}_{l+1}^{\text{RR}} - \mathcal{Q}_l^{\text{RR}}\| \leq 2\tilde{C}_A \alpha^2 (k - 1) (1 - \alpha a)^{(k-2)/2}.$$

*Proof of (i).* Take norms in the basic RR identity above. Using the equivalence  $\|\cdot\| \leq \kappa_Q^{1/2} \|\cdot\|_Q$  and the Lyapunov contraction at both step sizes,

$$\|2B_\alpha^k - B_{2\alpha}^k\|_Q \leq 2\|B_\alpha^k\|_Q + \|B_{2\alpha}^k\|_Q \leq 2(1 - \alpha a)^{k/2} + (1 - 2\alpha a)^{k/2} \leq 3(1 - \alpha a)^{k/2}.$$

Submultiplicativity in  $\|\cdot\|_Q$  followed by norm equivalence then gives

$$\|\mathcal{Q}_l^{\text{RR}} - \bar{A}^{-1}\| \leq \kappa_Q^{1/2} \|\bar{A}^{-1}\| \cdot \|2B_\alpha^k - B_{2\alpha}^k\|_Q \leq 3C_Q(1 - \alpha a)^{k/2}.$$

*Proof of (ii).* Apply the elementary identity  $X^m - Y^m = (X - Y) \sum_{i=1}^m X^{i-1} Y^{m-i}$  with  $X = B_\alpha$ ,  $Y = B_{2\alpha}$ ,  $X - Y = \alpha \bar{A}$ , and  $m = k - 1$ :

$$B_\alpha^{k-1} - B_{2\alpha}^{k-1} = \alpha \bar{A} \sum_{i=1}^{k-1} B_\alpha^{i-1} B_{2\alpha}^{k-1-i}.$$

Each summand obeys, by submultiplicativity in  $\|\cdot\|_Q$  and Lyapunov contraction,

$$\begin{aligned} \|B_\alpha^{i-1} B_{2\alpha}^{k-1-i}\| &\leq \kappa_Q^{1/2} (1 - \alpha a)^{(i-1)/2} (1 - 2\alpha a)^{(k-1-i)/2} \\ &\leq \kappa_Q^{1/2} (1 - \alpha a)^{(k-2)/2}, \end{aligned}$$

where in the last step we used  $1 - 2\alpha a \leq 1 - \alpha a$ . Summing over  $i \in \{1, \dots, k - 1\}$  and absorbing constants,

$$\|B_\alpha^{k-1} - B_{2\alpha}^{k-1}\| \leq \alpha \kappa_Q \|\bar{A}\| (k - 1) (1 - \alpha a)^{(k-2)/2}.$$

Inserting this bound into the discrete-difference identity finishes the proof.  $\square$

### 4.4. Summed Bounds and Comparison with the Single-Step Case

**Corollary 1.** Under the assumptions of the previous lemma, uniformly in  $n \geq 2$ ,

$$\sum_{l=1}^{n-1} \|Q_l^{\text{RR}} - \bar{A}^{-1}\|^2 \leq \frac{C_1}{\alpha a}, \quad \sum_{l=1}^{n-2} \|Q_{l+1}^{\text{RR}} - Q_l^{\text{RR}}\| \leq \frac{C_2}{a^2},$$

with  $C_1 = 9C_Q^2$  and  $C_2 = 32\tilde{C}_A$ .

*Proof.* For the first sum apply (i) and the geometric series  $\sum_{k \geq 1} (1 - \alpha a)^k \leq \frac{1}{\alpha a}$ . For the second, apply (ii) and

$$\sum_{k \geq 2} (k-1) (1 - \alpha a)^{(k-2)/2} = \sum_{m \geq 0} (m+1) (1 - \alpha a)^{m/2} \leq \frac{1}{(1 - \sqrt{1 - \alpha a})^2} \leq \frac{4}{(\alpha a)^2},$$

where the last step uses  $1 - \sqrt{1 - \alpha a} \geq \alpha \frac{a}{2}$  for  $\alpha a \leq 1/2$ . Multiplying by  $2\tilde{C}_A \alpha^2$  gives the claim, up to the stated universal constant.  $\square$

**Remark 2.** Comparison with the single-step case. The same identities for  $Q_l^{(\alpha)}$  give  $\|Q_l^{(\alpha)} - \bar{A}^{-1}\| \leq C_Q (1 - \alpha a)^{k/2}$  and  $\|Q_{l+1}^{(\alpha)} - Q_l^{(\alpha)}\| \leq \alpha \kappa_Q^{1/2} (1 - \alpha a)^{(k-1)/2}$ . Part (i) of the lemma above is therefore the same order as the single-step bound: the leading  $-\bar{A}^{-1}$  term of  $Q_{n-1}^{\text{RR}}$  is exactly  $-\bar{A}^{-1}$  (since  $Q_{n-1}^{\text{RR}} = 0$  identically), so no factor of  $\alpha$  is gained on the boundary. The Richardson–Romberg cancellation is visible in the successive difference of (ii): for fixed lag  $k$  the discrete derivative gains an extra factor of  $\alpha$ . This local gain comes with the factor  $k-1$ , so after summing over the geometric tail it should not be read as a global extra factor of  $\alpha$  in the Abel term. What is needed for the Poisson-equation remainder is the uniform total-variation bound in the corollary. On the variance side the corollary gives  $\sum \|Q_l^{\text{RR}} - \bar{A}^{-1}\|^2 = O(\frac{1}{\alpha a})$ , which yields the finite-horizon comparison error  $O(\frac{1}{n\alpha a})$ .

## 4.5. Variance Comparison

After the Poisson decomposition the leading martingale  $W^{\text{RR}}$  has predictable quadratic variation that converges to

$$\Sigma_n^{\text{RR}} := \frac{1}{n} \sum_{l=1}^{n-1} Q_l^{\text{RR}} \Sigma_\epsilon^{(\text{M})} (Q_l^{\text{RR}})^\top, \quad \sigma_n^{2, \text{RR}}(u) := u^\top \Sigma_n^{\text{RR}} u,$$

with  $\Sigma_\epsilon^{(\text{M})}$  the long-run noise covariance of the Markov chain,  $\Sigma_\epsilon^{(\text{M})} = \mathbb{E}_\pi [\epsilon(Z_0) \epsilon(Z_0)^\top] + 2 \sum_{j \geq 1} \mathbb{E}_\pi [\epsilon(Z_0) \epsilon(Z_j)^\top]$ . The asymptotic covariance is

$$\Sigma_\infty = \bar{A}^{-1} \Sigma_\epsilon^{(\text{M})} \bar{A}^{-\top}, \quad \sigma^2(u) = u^\top \Sigma_\infty u.$$

The bounds of the previous section give a quantitative comparison between  $\Sigma_n^{\text{RR}}$  and  $\Sigma_\infty$ .

**Lemma 8.** Let  $\alpha, 2\alpha \in (0, \alpha_\infty]$ , set  $\Sigma := \Sigma_\epsilon^{(\text{M})}$ , and assume  $\|\Sigma\| < \infty$ . Then

$$\|\Sigma_n^{\text{RR}} - \Sigma_\infty\| \leq \frac{C_3}{n\alpha a},$$

with  $C_3 = 12 C_Q \|\bar{A}^{-1}\| \|\Sigma\| + 9 C_Q^2 \|\Sigma\|$ . Consequently, for every  $u \in \mathbb{R}^d$ ,

$$|\sigma_n^{2, \text{RR}}(u) - \sigma^2(u)| \leq \frac{C_3 \|u\|^2}{n\alpha a}.$$

At the working scale  $\alpha = c n^{-1/2}$  this is  $O(n^{-1/2})$ .

*Proof.* Write  $\Delta_l := Q_l^{\text{RR}} - \bar{A}^{-1}$  and expand

$$Q_l^{\text{RR}} \Sigma (Q_l^{\text{RR}})^\top - \Sigma_\infty = \underbrace{\Delta_l \Sigma \bar{A}^{-\top} + \bar{A}^{-1} \Sigma \Delta_l^\top}_{R_{1,l}} + \underbrace{\Delta_l \Sigma \Delta_l^\top}_{R_{2,l}}.$$

Submultiplicativity of the operator norm yields the pointwise bounds

$$\| R_{1,l} \| \leq 2 \| \bar{A}^{-1} \| \| \Sigma \| \| \Delta_l \|, \quad \| R_{2,l} \| \leq \| \Sigma \| \| \Delta_l \|^2.$$

Summing the linear part with the bound from part (i) of the previous lemma and the geometric series  $\sum_{k \geq 1} (1 - \alpha a)^{k/2} \leq \frac{1}{1 - \sqrt{1 - \alpha a}} \leq \frac{2}{\alpha a}$  (valid for  $\alpha a \leq 1/2$ ),

$$\sum_{l=1}^{n-1} \| \Delta_l \| \leq 3C_Q \sum_{k=1}^{n-1} (1 - \alpha a)^{k/2} \leq \frac{6C_Q}{\alpha a}.$$

The quadratic part is bounded directly by the previous corollary,

$$\sum_{l=1}^{n-1} \| \Delta_l \|^2 \leq \frac{9C_Q^2}{\alpha a}.$$

Combining,

$$\begin{aligned} \| \Sigma_n^{\text{RR}} - \Sigma_\infty \| &\leq \frac{1}{n} \sum_{l=1}^{n-1} (\| R_{1,l} \| + \| R_{2,l} \|) \\ &\leq \frac{1}{n} \left( 2 \| \bar{A}^{-1} \| \| \Sigma \| \cdot \frac{6C_Q}{\alpha a} + \| \Sigma \| \cdot \frac{9C_Q^2}{\alpha a} \right) = \frac{C_3}{n \alpha a}, \end{aligned}$$

which proves the operator-norm bound. The scalar bound on  $|\sigma_n^{\text{RR}}(u) - \sigma^2(u)| = |u^\top (\Sigma_n^{\text{RR}} - \Sigma_\infty)u|$  follows from the Cauchy–Schwarz inequality.  $\square$

**Remark 3.** The constant  $C_3$  is independent of  $\alpha$  and  $n$ ; only  $a$  enters through the Lyapunov contraction. The bound is tight in the geometric scale: the dominant contribution comes from the near-boundary region  $l \approx n$ , where  $\| \Delta_l \|$  is bounded below by a positive constant (see the boundary identity  $Q_{n-1}^{\text{RR}} = 0$ ). The same calculation for the single-step PR weights yields  $\| \Sigma_n^{\text{PR}} - \Sigma_\infty \| \leq \frac{C}{n \alpha a}$ , so Richardson–Romberg does not improve the variance comparison rate, but it also does not degrade it: the leading-order variance is preserved.

## 4.6. Poisson Martingale Approximation

The variance comparison of the previous section identifies the limiting covariance of  $W^{\text{RR}}$ , but it does not by itself produce a martingale. This subsection converts  $W^{\text{RR}}$  into a martingale plus a quantitatively small remainder via the Poisson equation for the Markov chain  $\{Z_l\}$ . The deterministic kernel bounds of Sections 4.3–4.4 enter once and for all in the boundary/Abel control of the remainder.

**Poisson kernel.** Let  $Q$  denote the one-step Markov transition kernel of  $(Z_k)_{k \geq 1}$ , acting on bounded measurable functions  $f : Z \rightarrow \mathbb{R}^d$  by

$$(Qf)(z) = \mathbb{E}[f(Z_{k+1}) \mid Z_k = z].$$

Under UGE 1 with mixing time  $t_{\text{mix}}$ , the geometric Dobrushin bound  $\| Q^k \epsilon \|_\infty \leq 2 \| \epsilon \|_\infty (1/4)^{\lfloor k/t_{\text{mix}} \rfloor}$  (valid for centered  $\epsilon$ ,  $\pi(\epsilon) = 0$ ) makes the Poisson series

$$\hat{\epsilon} := \sum_{k=0}^{\infty} Q^k \epsilon$$

absolutely convergent in sup-norm, with

$$\| \hat{\epsilon} \|_\infty \leq 2 \| \epsilon \|_\infty \sum_{k=0}^{\infty} (1/4)^{\lfloor k/t_{\text{mix}} \rfloor} \leq 3 t_{\text{mix}} \| \epsilon \|_\infty, \quad \| Q \hat{\epsilon} \|_\infty \leq \| \hat{\epsilon} \|_\infty,$$

the second inequality because  $Q$  is a Markov kernel and contracts the sup-norm. By construction  $\hat{\epsilon}$  solves the **Poisson equation**

$$\hat{\epsilon} - Q\hat{\epsilon} = \epsilon.$$

**Conditional centering.** Let  $\mathcal{F}_l := \sigma(Z_1, \dots, Z_l)$ . Substituting into the noise sample and adding and subtracting the conditional mean,

$$\epsilon(Z_l) = \underbrace{[\hat{\epsilon}(Z_l) - Q\hat{\epsilon}(Z_{l-1})]}_{\text{martingale increment}} + \underbrace{[Q\hat{\epsilon}(Z_{l-1}) - Q\hat{\epsilon}(Z_l)]}_{\text{telescope}}, \quad l \geq 2,$$

where the first bracket is centered conditionally on  $\mathcal{F}_{l-1}$  since the Markov property gives  $\mathbb{E}[\hat{\epsilon}(Z_l) \mid \mathcal{F}_{l-1}] = Q\hat{\epsilon}(Z_{l-1})$ . The  $l = 1$  term has no past state to condition on; we treat it directly via  $\epsilon(Z_1) = \hat{\epsilon}(Z_1) - Q\hat{\epsilon}(Z_1)$ . The decomposition therefore separates  $W^{\text{RR}}$  into a martingale piece and a deterministic-coefficient telescope; Abel summation against the kernel sequence  $\{Q_l^{\text{RR}}\}$  converts the telescope into boundary terms plus a sum against the discrete derivative  $Q_{l+1}^{\text{RR}} - Q_l^{\text{RR}}$ , which is exactly the object bounded in the Corollary of Section 4.4.

**Lemma 9.** Assume **UGE 1** and  $\pi(\epsilon) = 0$ . Set

$$\Delta M_l^{\text{RR}} := Q_l^{\text{RR}} (\hat{\epsilon}(Z_l) - Q\hat{\epsilon}(Z_{l-1})), \quad 2 \leq l \leq n-1,$$

and let  $M_n^{\text{RR}} := \sum_{l=2}^{n-1} \Delta M_l^{\text{RR}}$ . Then  $\{\Delta M_l^{\text{RR}}\}_{l=2}^{n-1}$  is a sequence of  $\mathcal{F}_l$ -martingale differences, and

$$W^{\text{RR}} = -\frac{1}{\sqrt{n}} M_n^{\text{RR}} + D_{2,n}^{\text{RR}},$$

with the **Poisson boundary/Abel remainder**

$$D_{2,n}^{\text{RR}} := -\frac{1}{\sqrt{n}} \left[ Q_1^{\text{RR}} \hat{\epsilon}(Z_1) + \sum_{l=1}^{n-2} (Q_{l+1}^{\text{RR}} - Q_l^{\text{RR}}) Q\hat{\epsilon}(Z_l) \right].$$

(The right boundary  $Q_{n-1}^{\text{RR}} Q\hat{\epsilon}(Z_{n-1})$  vanishes identically because  $Q_{n-1}^{\text{RR}} = 0$ .) Moreover, with  $C_Q := \|\bar{A}^{-1}\| + 3C_Q$  a uniform bound on  $\|Q_l^{\text{RR}}\|$ , and  $C_2$  the constant from the Corollary of Section 4.4,

$$\|D_{2,n}^{\text{RR}}\|_{\infty} \leq \frac{3t_{\text{mix}} \|\epsilon\|_{\infty}}{\sqrt{n}} \left( C_Q + \frac{C_2}{a^2} \right).$$

Consequently, for every  $p \geq 1$ ,

$$\|D_{2,n}^{\text{RR}}\|_{L_p} \leq \frac{C t_{\text{mix}} \|\epsilon\|_{\infty}}{a^2 \sqrt{n}},$$

with a constant  $C$  depending only on  $\|\bar{A}^{-1}\|$ ,  $\kappa_Q$ , and  $\|\bar{A}\|$ .

*Proof.* The increments  $\Delta M_l^{\text{RR}}$  are  $\mathcal{F}_l$ -martingale differences because  $Q_l^{\text{RR}}$  is deterministic and the Markov property gives  $\mathbb{E}[\hat{\epsilon}(Z_l) \mid \mathcal{F}_{l-1}] = Q\hat{\epsilon}(Z_{l-1})$ .

For the decomposition, substitute into the definition :

$$W^{\text{RR}} = -\frac{1}{\sqrt{n}} \sum_{l=1}^{n-1} Q_l^{\text{RR}} (\hat{\epsilon}(Z_l) - Q\hat{\epsilon}(Z_l)).$$

Peel off  $l = 1$  and apply the conditional-centering identity above to the remaining  $l \in \{2, \dots, n-1\}$ :

$$W^{\text{RR}} = -\frac{1}{\sqrt{n}} Q_1^{\text{RR}} \hat{\epsilon}(Z_1) + \frac{1}{\sqrt{n}} Q_1^{\text{RR}} Q\hat{\epsilon}(Z_1) - \frac{1}{\sqrt{n}} M_n^{\text{RR}} - \frac{1}{\sqrt{n}} \sum_{l=2}^{n-1} Q_l^{\text{RR}} (Q\hat{\epsilon}(Z_{l-1}) - Q\hat{\epsilon}(Z_l)).$$

Set  $g_l := Q\hat{\epsilon}(Z_l)$  and Abel-sum the last sum:

$$\sum_{l=2}^{n-1} Q_l^{\text{RR}}(g_{l-1} - g_l) = Q_2^{\text{RR}} g_1 - Q_{n-1}^{\text{RR}} g_{n-1} + \sum_{l=2}^{n-2} (Q_{l+1}^{\text{RR}} - Q_l^{\text{RR}}) g_l.$$

Adding the contribution  $Q_1^{\text{RR}} g_1$  from the peeled  $l = 1$  term combines with  $-Q_2^{\text{RR}} g_1$  to yield  $-(Q_2^{\text{RR}} - Q_1^{\text{RR}}) g_1$ , which folds the boundary  $l = 1$  summand back into the discrete-derivative sum:

$$\begin{aligned} & -Q_1^{\text{RR}} \hat{\epsilon}(Z_1) + Q_1^{\text{RR}} g_1 - Q_2^{\text{RR}} g_1 + Q_{n-1}^{\text{RR}} g_{n-1} - \sum_{l=2}^{n-2} (Q_{l+1}^{\text{RR}} - Q_l^{\text{RR}}) g_l \\ & = -Q_1^{\text{RR}} \hat{\epsilon}(Z_1) - \sum_{l=1}^{n-2} (Q_{l+1}^{\text{RR}} - Q_l^{\text{RR}}) g_l + Q_{n-1}^{\text{RR}} g_{n-1}. \end{aligned}$$

The rightmost boundary  $Q_{n-1}^{\text{RR}} g_{n-1}$  vanishes because  $Q_{n-1}^{\text{RR}} = 2\alpha I - 2\alpha I = 0$  identically (from  $Q_{n-1}^{(\alpha)} = \alpha I$  and  $Q_{n-1}^{(2\alpha)} = 2\alpha I$ ). Multiplying by  $1/\sqrt{n}$  and absorbing the sign into  $D_{2,n}^{\text{RR}}$  produces exactly the stated formula.

For the sup-norm bound, use  $\|g_l\|_\infty \leq \|\hat{\epsilon}\|_\infty \leq 3t_{\text{mix}} \|\epsilon\|_\infty$ , the uniform bound  $\|Q_1^{\text{RR}}\| \leq C_Q$  on the left boundary, and the summed-total-variation bound  $\sum_{l=1}^{n-2} \|Q_{l+1}^{\text{RR}} - Q_l^{\text{RR}}\| \leq C_2/a^2$  on the Abel sum:

$$\|\sqrt{n} D_{2,n}^{\text{RR}}\| \leq 3t_{\text{mix}} \|\epsilon\|_\infty (C_Q + C_2/a^2).$$

The  $L_p$  bound is immediate from the deterministic sup-norm bound.  $\square$

**Remark 4. Two structural features.** First, the bound is **deterministic**: no martingale concentration or moment inequality is invoked, only pointwise bounds on  $Q_l^{\text{RR}}$ , on its discrete derivative, and on  $\hat{\epsilon}$ . The dependence on  $p$  promised in the working plan ( $\text{poly}(p, t_{\text{mix}})$ ) is therefore absorbed into the constant; the effective scaling is  $\|D_{2,n}^{\text{RR}}\|_{L_p} \leq C t_{\text{mix}}/(a^2 \sqrt{n})$ . Second, the  $1/a^2$  factor is **exactly** the cost of the discrete derivative summed Abel-style: it is the same  $C_2/a^2$  that appears in the second part of the Corollary of Section 4.4. No further  $1/\alpha$  blow-up enters, because the kernel-difference bound of Section 4.3 already carries the cancelling  $\alpha^2(k-1)$  prefactor.

**Boundary asymmetry.** The right-end boundary vanishes because  $Q_{n-1}^{\text{RR}} = 0$  — a Richardson–Romberg cancellation that does not occur for the single-step PR weight  $Q_{n-1}^{(\alpha)} = \alpha I$ , which contributes a (small)  $\alpha t_{\text{mix}}/\sqrt{n}$  residue. RR therefore **removes** one of the two Poisson-boundary terms outright.

**Comparison with martingale scale.** The leading martingale piece  $M_n^{\text{RR}}/\sqrt{n}$  has  $L_2$ -norm of order 1 (it is the natural  $\sqrt{n}$ -scaled object whose CLT variance equals  $\sigma_n^{2, \text{RR}}(u)$  at leading order, by the variance-comparison lemma). The remainder  $D_{2,n}^{\text{RR}}$  is therefore  $O(n^{-1/2})$  relative to it, well below the target Berry–Esseen scale  $n^{-1/4}$ .

## 4.7. Predictable Quadratic Variation Concentration

The Berry–Esseen step for the martingale  $M_n^{\text{RR}}$  requires a quantitative control of its predictable quadratic variation  $\langle M^{\text{RR}} \rangle_n$  around its deterministic asymptotic counterpart  $n \sigma_n^{2, \text{RR}}(u)$ . This subsection produces such a control by applying the Markov concentration result of Section 2.2 to a suitable centered quadratic functional of the chain. The result is the RR analogue of Lemmas 22–23 of Samsonov et al. (2025).

**Predictable quadratic variation.** By the Poisson Martingale Approximation lemma above, the increments  $\Delta M_l^{\text{RR}} = Q_l^{\text{RR}}(\hat{\epsilon}(Z_l) - Q\hat{\epsilon}(Z_{l-1}))$  are  $\mathcal{F}_l$ -martingale differences. The Markov property  $\mathbb{E}[\hat{\epsilon}(Z_l) \mid \mathcal{F}_{l-1}] = Q\hat{\epsilon}(Z_{l-1})$  gives, by direct computation,

$$\begin{aligned} \mathbb{E}[\Delta M_l^{\text{RR}} (\Delta M_l^{\text{RR}})^\top \mid \mathcal{F}_{l-1}] &= Q_l^{\text{RR}} \mathbb{E}[\hat{\epsilon}(Z_l) \hat{\epsilon}(Z_l)^\top \mid \mathcal{F}_{l-1}] (Q_l^{\text{RR}})^\top \\ &\quad - Q_l^{\text{RR}} Q\hat{\epsilon}(Z_{l-1}) (Q\hat{\epsilon}(Z_{l-1}))^\top (Q_l^{\text{RR}})^\top \\ &= Q_l^{\text{RR}} \bar{\epsilon}(Z_{l-1}) (Q_l^{\text{RR}})^\top, \end{aligned}$$



where the cross-terms in the expansion of  $(\hat{\epsilon}(Z_l) - Q\hat{\epsilon}(Z_{l-1}))(\hat{\epsilon}(Z_l) - Q\hat{\epsilon}(Z_{l-1}))^\top$  cancel by the Markov property and we have set

$$\bar{\epsilon}(z) := Q(\hat{\epsilon}\hat{\epsilon}^\top)(z) - (Q\hat{\epsilon})(z)(Q\hat{\epsilon})(z)^\top.$$

Summing over  $l \in \{2, \dots, n-1\}$  and tracking that  $\langle M^{\text{RR}} \rangle_n := \sum_{l=2}^{n-1} \mathbb{E}[\Delta M_l^{\text{RR}} (\Delta M_l^{\text{RR}})^\top \mid \mathcal{F}_{l-1}]$ ,

$$\langle M^{\text{RR}} \rangle_n = \sum_{l=2}^{n-1} Q_l^{\text{RR}} \bar{\epsilon}(Z_{l-1}) (Q_l^{\text{RR}})^\top.$$

The function  $\bar{\epsilon}$  has stationary mean equal to the long-run noise covariance,

$$\pi(\bar{\epsilon}) = \Sigma_\epsilon^{(\text{M})} =: \Sigma,$$

which is a standard identity for the Poisson solution of a Markov chain (Samsonov et al. 2025, Eq. (10); see also Douc–Moulines–Priouret–Soulier 2018, Theorem 21.2.5).

**Sup-norm bound on  $\bar{\epsilon}$ .** Since  $\|\hat{\epsilon}\|_\infty \leq 3t_{\text{mix}}\|\epsilon\|_\infty$  and  $Q$  is a Markov kernel (so  $Qf$  inherits the sup-norm of  $f$ ),

$$\begin{aligned} \|\bar{\epsilon}\|_\infty &\leq \|Q(\hat{\epsilon}\hat{\epsilon}^\top)\|_\infty + \|(Q\hat{\epsilon})(Q\hat{\epsilon})^\top\|_\infty \\ &\leq 2\|\hat{\epsilon}\|_\infty^2 \leq 18t_{\text{mix}}^2\|\epsilon\|_\infty^2. \end{aligned}$$

**Lemma 10.** Assume **UGE 1** and  $\pi(\epsilon) = 0$ , with  $\|\epsilon\|_\infty < \infty$ . Let  $C_Q$  be the uniform bound on  $\|Q_l^{\text{RR}}\|$  from the previous lemma. There exists a universal constant  $C_4 > 0$  such that, for every  $u \in \mathbb{R}^d$ , every  $p \geq 2$ , every initial distribution  $\xi$ , and every  $n \geq 2$ ,

$$\mathbb{E}_\xi^{1/p}[\|u^\top \langle M^{\text{RR}} \rangle_n u - n\sigma_n^{2, \text{RR}}(u)\|^p] \leq C_4 C_Q^2 \|u\|^2 \|\epsilon\|_\infty^2 t_{\text{mix}}^{5/2} \sqrt{pn}.$$

*Proof.* Write  $h_{l(z)} := u^\top Q_l^{\text{RR}} \bar{\epsilon}(z) (Q_l^{\text{RR}})^\top u$  and  $g_{l(z)} := h_{l(z)} - \pi(h_l)$ . Submultiplicativity of the operator norm and give

$$|h_{l(z)}| \leq C_Q^2 \|u\|^2 \|\bar{\epsilon}\|_\infty \leq 18C_Q^2 \|u\|^2 t_{\text{mix}}^2 \|\epsilon\|_\infty^2, \quad \|g_l\|_\infty \leq 2|h_{l(z)}| \leq 36C_Q^2 \|u\|^2 t_{\text{mix}}^2 \|\epsilon\|_\infty^2.$$

By and the variance definition  $n\sigma_n^{2, \text{RR}}(u) = \sum_{l=1}^{n-1} \pi(h_l)$ ,

$$u^\top \langle M^{\text{RR}} \rangle_n u - n\sigma_n^{2, \text{RR}}(u) = \sum_{l=2}^{n-1} g_{l(Z_{l-1})} - \pi(h_1).$$

The boundary scalar is deterministic and obeys  $|\pi(h_1)| \leq C_Q^2 \|u\|^2 \|\Sigma\| \leq 18C_Q^2 \|u\|^2 t_{\text{mix}}^2 \|\epsilon\|_\infty^2$ , where the last step uses  $\|\Sigma\| = \|\pi(\bar{\epsilon})\| \leq \|\bar{\epsilon}\|_\infty$  and .

For the centered sum, set  $\tilde{g}_i := g_{i+1}$  for  $i \in \{1, \dots, n-2\}$ . By construction  $\pi(\tilde{g}_i) = 0$  and  $\|\tilde{g}_i\|_\infty \leq c := 36C_Q^2 \|u\|^2 t_{\text{mix}}^2 \|\epsilon\|_\infty^2$ . The Markov concentration lemma of Section 2.2 (Levin et al. 2025, Lemma 11) applied to  $\sum_{i=1}^{n-2} \tilde{g}_{i(Z_i)}$  yields, for every  $t \geq 0$ ,

$$\mathbb{P}_\xi \left( \left| \sum_{i=1}^{n-2} \tilde{g}_{i(Z_i)} \right| \geq t \right) \leq 2 \exp \left( -\frac{t^2}{2u_n^2} \right), \quad u_n^2 \leq 64t_{\text{mix}}(n-2)c^2.$$

The sub-Gaussian-to-moment lemma of Section 2.2 then gives, for  $p \geq 2$ ,

$$\mathbb{E}_\xi^{1/p} \left[ \left| \sum_{l=2}^{n-1} g_{l(Z_{l-1})} \right|^p \right] \leq 2^{1/p} \sqrt{p} u_n \leq 2\sqrt{p} u_n \leq 16\sqrt{2} \sqrt{p(n-2)} c \sqrt{t_{\text{mix}}}.$$

Substituting  $c$  and bounding  $\sqrt{n-2} \leq \sqrt{n}$ ,

$$\mathbb{E}_\xi^{1/p} \left[ \left| \sum_{l=2}^{n-1} g_{l(Z_{l-1})} \right|^p \right] \leq 16\sqrt{2} \cdot 36 C_Q^2 \|u\|^2 \|\epsilon\|_\infty^2 t_{\text{mix}}^{5/2} \sqrt{pn} = 576\sqrt{2} C_Q^2 \|u\|^2 \|\epsilon\|_\infty^2 t_{\text{mix}}^{5/2} \sqrt{pn}.$$

The boundary contribution satisfies  $|\pi(h_1)| \leq 18 C_Q^2 \|u\|^2 t_{\text{mix}}^2 \|\epsilon\|_\infty^2 \leq 18 C_Q^2 \|u\|^2 t_{\text{mix}}^{5/2} \|\epsilon\|_\infty^2 \sqrt{pn}$  for every  $p, n \geq 1$  (using  $t_{\text{mix}} \geq 1$  and  $\sqrt{pn} \geq 1$ ). Adding the two pieces in by the triangle inequality and rounding the resulting prefactor to  $C_4 := 850$  gives  $\square$ .

**Corollary 2.** Under the assumptions of the previous lemma, for every  $u \in \mathbb{R}^d$ , every  $p \geq 2$ , and every  $n \geq 2$ ,

$$\mathbb{E}_\xi^{1/p} [ |u^\top \langle M^{\text{RR}} \rangle_n u - n \sigma^2(u)|^p ] \leq C_4 C_Q^2 \|u\|^2 \|\epsilon\|_\infty^2 t_{\text{mix}}^{5/2} \sqrt{pn} + \frac{C_3 \|u\|^2}{\alpha a},$$

with  $C_3$  the variance-comparison constant of Section 4.5.

*Proof.* The triangle inequality and the variance-comparison lemma give

$$|u^\top \langle M^{\text{RR}} \rangle_n u - n \sigma^2(u)| \leq |u^\top \langle M^{\text{RR}} \rangle_n u - n \sigma_n^{2, \text{RR}}(u)| + n |\sigma_n^{2, \text{RR}}(u) - \sigma^2(u)|.$$

The first piece is bounded in  $L_p$  by ; the second is the deterministic bound  $n C_3 \|u\|^2 / (n \alpha a) = C_3 \|u\|^2 / (\alpha a)$ .  $\square$

**Remark 5. Working scale and Berry–Esseen consequence.** The bound  $\|u^\top \langle M^{\text{RR}} \rangle_n u - n \sigma_n^{2, \text{RR}}(u)\|_{L_p} \leq C t_{\text{mix}}^{5/2} \sqrt{pn}$  is exactly what feeds into the martingale Berry–Esseen of Bolthausen–Fan type (cf. Lemma 21 / Proposition 13 of Samsonov et al. 2025): the conditional-variance term in the Bolthausen–Fan inequality is  $\sqrt{p} s_n^{-2p/(2p+1)} \mathbb{E}^{1/(2p+1)} [\|\sum \sigma_l^2 - s_n^2\|^p]$ , which with  $s_n^2 = n \sigma_n^{2, \text{RR}}(u) \asymp n$  (uniformly in  $\alpha$  at the working scale) and the exponent  $\sqrt{pn} \rightarrow n^{p/(2p+1)}$  gives the standard  $n^{-1/4}$  rate after taking  $p = \lceil \log n \rceil$ .

**RR vs single-step structure.** The bound is structurally identical to the single-step PR case (Samsonov et al. 2025, Lemma 23): the only RR-specific input is the uniform sup bound  $\|Q_l^{\text{RR}}\| \leq C_Q$ . No extra  $1/a$  or  $1/\alpha$  factor enters, because the Markov concentration is applied to a **bounded** function and the weights  $Q_l^{\text{RR}}$  enter only through their sup-norm. Variance comparison (Section 4.5) is the only place where  $1/(\alpha a)$  appears, and at the working scale  $\alpha = c n^{-1/2}$  this contributes  $O(\sqrt{n})$  — the same order as the leading concentration term.

**Why we centred against  $\sigma_n^{2, \text{RR}}(u)$ .** The natural normalisation in the Bolthausen–Fan martingale BE step is  $s_n^2 = \sum_l \sigma_l^2$ , which is precisely  $n \sigma_n^{2, \text{RR}}(u)$  for the increments  $\Delta M_l^{\text{RR}}$ . Centering against  $n \sigma^2(u)$  (corollary above) is used in the corollary that converts the BE statement from the  $\sigma_n^{\text{RR}}(u)$  normalisation to the asymptotic  $\sigma(u)$  normalisation.